MIL-M-38510/52D
30 April 1984
SUPERSEDING
MIL-M-38510/52C
7 December 1979

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, NOR GATES, MONOLITHIC SILICON, POSITIVE LOGIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS, NOR gate logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Dual-3-input NOR gate (plus inverter)
02	Quad 2-input NOR gate
03	Dual 4-input NOR gate
04	Triple 3-input NOR gate
51	Dual 3-input NOR gate (plus inverter)
52	Quad 2-input NOR gate
53	Dual 4-input NOR gate
54	Triple 3-input NOR gate

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
X	F-1 (14-lead, 1/4" x 1/4"), flat package, except A dimension = 0.1" (2.54 mm) max.
Y	F-2 (14-lead, 1/4" x 3/8"), flat package, except A dimension = 0.1" (2.54 mm) max.

NOTES:

1. As an exception to nickel plate or undercoating paragraph of MIL-M-38510, for case outlines X and Y only, the leads of bottom brazed ceramic packages (i.e., configuration 2 of case outlines F-1 or F-2) may have electroless nickel undercoating which shall be 50 to 200 microinches (1.27 to 5.08 µm) thick provided the lead finish is hot solder dip (i.e., finish letter A) and provided that, after any lead forming, an additional hot solder dip coating is applied which shall extend from the outer tip of the lead to no more than 0.015 inch (0.38 mm) from the package edge.

2. For bottom or side brazed packages, case outlines X and Y only, the S_1 dimension may go to .000 inch (.00 mm) minimum.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: George C. Marshal Space Flight Center, ATTN: EG02, Marshall Space Flight Center, Alabama 35812 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage range (V_{DD}-V_{SS}):	
Device types 01, 02, 03, and 04	-0.5 V dc to +15.5 V dc
Device types 51, 52, 53, and 54	-0.5 V dc to +18.0 V dc
Input current (each input)	±10 mA
Input voltage range	(V_{SS}-0.5 V dc) ≤V_I ≤(V_{DD}+0.5 V dc)
Storage temperature range	-65°C to +175°C
Maximum power dissipation (P_D)	200 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case	(See MIL-M-38510, appendix C)
Junction temperature (T_J)	+175°C

1.4 Recommended operating conditions.

Supply voltage range (V_{DD}-V_{SS}):	
Device types 01, 02, 03, and 04	4.5 V dc to 12.5 V dc
Device types 51, 52, 53, and 54	4.5 V dc to 15 V dc
Input low voltage range (V_{IL}):	
Device types 01, 02, 03, and 04	0 to 0.85 V dc @ V_{DD} = 5 V dc; 0 to 2.0 V dc @ V_{DD} = 10 V dc; 0 to 2.1 V dc @ V_{DD} = 12.5 V dc;
Device types 51, 52, 53, and 54	0 to 1.5 V dc @ V_{DD} = 5 V dc; V_{OL} = 10% V_{DD}; V_{OH} = 90% V_{DD}; 0 to 2.0 V dc @ V_{DD} = 10 V dc; 0 to 4.0 V dc @ V_{DD} = 15 V dc
Input high voltage range (V_{IH}):	
Device types 01, 02, 03, and 04	3.95 to 5.0 V dc @ V_{DD} = 5 V dc; 8 to 10 V dc @ V_{DD} = 10 V dc; 10.0 to 12.5 V dc @ V_{DD} = 12.5 V dc;
Device types 51, 52, 53, and 54	3.5 to 5.0 V dc V_{DD} = 5 V dc; V_{OL} = 10% V_{DD}; V_{OH} = 90% V_{DD}; 8 to 10.0 V dc @ V_{DD} = 10 V dc; 11.0 to 15.0 V dc @ V_{DD} = 15 V dc
Ambient operating temperature range (T_A)	-55°C to +125°C
Load capacitance	50 pF maximum

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

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3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein. Although eutectic die bonding is preferred, epoxy die bonding may be performed. However, the resin used shall be Dupont 5504 Conductive Silver Paste, or equivalent, which is cured at 200°C ±10°C for a minimum of 2 hours. The use of equivalent epoxies or cure cycles shall be approved by the qualifying activity. Equivalency shall be demonstrated in data submitted to the qualifying activity for verification.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1. Total dose radiation hardness identifier. Total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and 4.5.5 herein.

3.6.2 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.3 Correctness of indexing and marking. All devices shall be subjected to the final electrical tests specified in table II after part number marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 36 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in 3.1.9 through 3.1.13 of method 5004 and substitute lines 1 through 7 of table II herein.

b. Burn-in (method 1015 of MIL-STD-883).

(1) Static tests (test condition A) using circuit shown on figure 3, or equivalent. Ambient temperature shall be 125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

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(2) Dynamic test (test condition D) using circuit shown on figure 4, or equivalent. Ambient temperature shall be 125°C minimum. Test duration shall be in accordance with table I of method 1015.

c. Interim and final electrical parameters shall be as specified in table II herein.

d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failures shall be cumulative for determining PDA.

c. The PDA for class B devices shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics, after burn-in, exceed the specified delta limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.3.1 Qualification extension. When authorized by the qualifying activity, if a manufacturer qualifies to a 51-54 device type which is manufactured identically to a 01-04 device type on this specification, then the 01-04 device type may be part I qualified by conducting only group A electrical tests and any electrical tests specified as additional group C subgroups and by submitting data in accordance with MIL-M-38510, appendix D (i.e. groups A, B, C, D, and E tests are not required).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5, 6, 7, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_I measurement) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady state life (accelerated) test circuits shall be submitted to the qualifying activity for approval. When the alternate steady state life test is used, the circuit on figure 4, or equivalent, shall be used.

Text continues on page 52.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions V_{SS} = 0 V -55°C ≤ T_A ≤ 125°C unless otherwise specified	Device	Limits Min	Limits Max	Units
Positive clamping Input to V_{DD}	$V_{IC(pos)}$	T_A = 25°C V_{DD} = GND, V_{SS} = Open, Output = Open I_I = 1 mA	All		1.5	Vdc
Negative clamping Input to V_{SS}	$V_{IC(neg)}$	T_A = 25°C V_{DD} = Open, V_{SS} = GND, Output = Open I_I = -1 mA			-6	Vdc
Quiescent supply current	I_{SS}	Any combination of Inputs; V_{DD} = 15 Vdc	01, 02 03, 04		-750	nA
		V_{DD} = 18 Vdc	51, 52 53, 54			
High level output voltage	V_{OH1}	V_{DD} = 4.5 Vdc, I_{OH} = -.1 mA All inputs = V_{IL} (see table III notes)	01, 02 03, 04	2.5		Vdc
	V_{OH2}	V_{DD} = 5 Vdc, I_{OH} = -.13 mA All inputs = V_{IL} (see table III notes)	01, 02 03, 04	4.2		Vdc
	V_{OH3}	V_{DD} = 5 Vdc, I_{OH} = 0 All inputs = V_{IL} (see table III notes)	01, 02 03, 04	4.95		Vdc
	V_{OH4}	V_{DD} = 12.5 Vdc, I_{OH} = 0 All inputs = V_{IL} (see table III notes)	01, 02 03, 04	11.25		Vdc
	V_{OH5}	V_{DD} = 15 Vdc, I_{OH} = 0 All inputs = GND	51, 52 53, 54	14.95		Vdc
Low level output voltage	V_{OL1}	V_{DD} = 5.5 Vdc, I_{OL} = .23 mA Any one input = V_{IH} (see table III notes)	01, 02 03, 04		0.5	Vdc
	V_{OL2}	V_{DD} = 5 Vdc, I_{OL} = .28 mA Any one input = V_{IH} (see table III notes)	01, 02 03, 04		0.7	Vdc
	V_{OL3}	V_{DD} = 5 Vdc, I_{OL} = 0 Any one input = V_{IH} (see table III notes)	01, 02 03, 04		50	mV
	V_{OL4}	V_{DD} = 12.5 Vdc, I_{OL} = 0 Any one input = V_{IH} (see table III notes)	01, 02 03, 04		1.25	Vdc
	V_{OL5}	V_{DD} = 5 Vdc, I_{OL} = .56 mA All inputs = V_{IH} (see table III notes and 4.4.1c.)	01, 02 03, 04		0.5	Vdc
	V_{OL6}	V_{DD} = 15 Vdc I_{OL} = 0 Input = 15 V	51, 52 53, 54		0.05	Vdc
Input high voltage	V_{IH1}	V_{DD} = 5 Vdc Any one input = 3.5 Vdc. All other inputs = 1.5 Vdc V_O = 0.5 V $\|I_O\| \leq 1$ µA	51, 52 53, 54	3.5		Vdc
	V_{IH2}	V_{DD} = 10 Vdc Any one input = 7.0 V. All other inputs = 3.0 V V_O = 1.0 V $\|I_O\| \leq 1$ µA	51, 52 53, 54	7.0		Vdc
	V_{IH3}	V_{DD} = 15 Vdc Any one input = 11 V. All other inputs = 4.0 V $\|I_O\| \leq 1$ µA	51, 52 53, 54	11.0		Vdc

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TABLE I. Electrical performance characteristics – Continued.

Test	Symbol	Conditions V_{SS} = 0 V −55°C ≤ T_A ≤ 125°C unless otherwise specified		Device	Limits Min	Limits Max	Units
Input low voltage	V_{IL1}	V_{DD} = 5 Vdc All inputs = 1.5 Vdc V_O = 4.5 V $\|I_O\| \leq 1\ \mu A$		51, 52 53, 54		1.5	Vdc
	V_{IL2}	V_{DD} = 10 Vdc All inputs = 3.0 Vdc V_O = 9.0 V $\|I_O\| \leq 1\ \mu A$		51, 52 53, 54		3.0	Vdc
	V_{IL3}	V_{DD} = 15 Vdc All inputs = 4.0 Vdc V_O = 13.5 V $\|I_O\| \leq 1 \mu A$		51, 52 53, 54		4.0	Vdc
Output low (sink) current	I_{OL1}	V_{DD} = 5 Vdc V_{IN} = 5 Vdc V_{OL} = 0.4 Vdc		51, 52 53, 54	0.36		mAdc
	I_{OL2}	V_{DD} = 15 Vdc V_{IN} = 15 Vd V_{OL} = 1.5 Vdc		51, 52 53, 54	2.4		mAdc
Output high (source) current	I_{OH1}	V_{DD} = 5 Vdc All inputs = GND V_{OH} = 4.6 V		51, 52 53, 54	-0.36		mAdc
	I_{OH2}	V_{DD} = 15 Vdc All inputs = GND V_{OH} = 13.5 Vdc		51, 52 53, 54	-2.4	–	mAdc
Input leakage current 1/	I_{IH}	Measure inputs sequentially	V_{DD} = 15 Vdc	01, 02 03, 04		45	nA
			V_{DD} = 18 Vdc	51, 52 53, 54			
	I_{IL}	Measure inputs sequentially	V_{DD} = 15 Vdc	01, 02 03, 04		-45	nA
			V_{DD} = 18 Vdc	51, 52 53, 54			
Input test voltage	V_{ZAP}	(C_1 = 100 pF R_Z = 1.5 kΩ (see 4.5.3)		All	400		V
Input capacitance	C_I	V_{DD} = 0 Vdc, f = 1 MHz T_A = 25°C		All		12	pF
Propagation delay times High to low level	t_{PHL}	V_{DD} = 5 Vdc, CL = 50 pF see figure 5		All	11	315	ns
Low to high level	t_{PLH}			01, 51	14	415	ns
				02, 03, 04 52, 53, 54		315	
Transition times High to low level	t_{THL}	V_{DD} = 5 Vdc, CL = 50 pF see figure 5		All	15	450	ns
Low to high level	t_{TLH}			01, 51	23	675	ns
				02, 03, 04 52, 53, 54	21	615	

1/ input current of one input node.

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TABLE II. Burn-in and electrical test requirements.

Line no.	Applicable tests and MIL-STD-883 test methods	Class S devices 1/				Class B devices 1/			
		Ref. par.	Table 2/ III sub-groups	Table 3/ IV delta limits	Test circuit figure	Ref. par.	Table 2/ III sub-groups	Table 3/ IV delta limits	Test circuit figure
1	Interim electrical parameters (method 5004)		1				1		
2	Static burn-in I (method 1015)	4.2b 4.5.2			3				
3	Same as line 1		1	Δ					
4	Static burn-in II (method 1015)	4.2b 4.5.2			3	4.2b 4.5.2	4/		3
5	Same as line 1	4.2.d	1*	Δ		4.2d	1*	Δ	
6	Dynamic burn-in (method 1015)	4.2b 4.5.2			4				
7	Same as line 1	4.2	1*	Δ					
8	Final electrical parameters (method 5004)		1*,2,3,9				1*,2,3,9		
9	Group A end-point electrical parameters (method 5005)	4.4.1	1,2,3,4, 9,10,11			4.4.1	1,2,3, 4,9		
10	Group B end-point parameters (method 5005)	4.4.2	1,2,3,9, 10,11	Δ					
11	Group C end-point electrical parameters (method 5005)					4.4.3c	1,2,3	Δ	
12	Additional group C tests, (method 5005)					4.4.3	10,11		
13	Group D end-point electrical parameters (method 5005)	4.4.4	1,2,3			4.4.4	1,2,3		

1/ Blank spaces indicate tests are not applicable.

2/ (*) indicated PDA applied to subgroup 1 (see 4.2.1).

3/ (Δ) indicates delta limit shall be required in table III subgroup 1, where specified, and delta values shall be computed with reference to the previous interim electrical parameters.

4/ The device manufacturer may at his option either perform delta measurements or within 24 hours after burn-in (or removal of bias), perform the final electrical parameter measurements.

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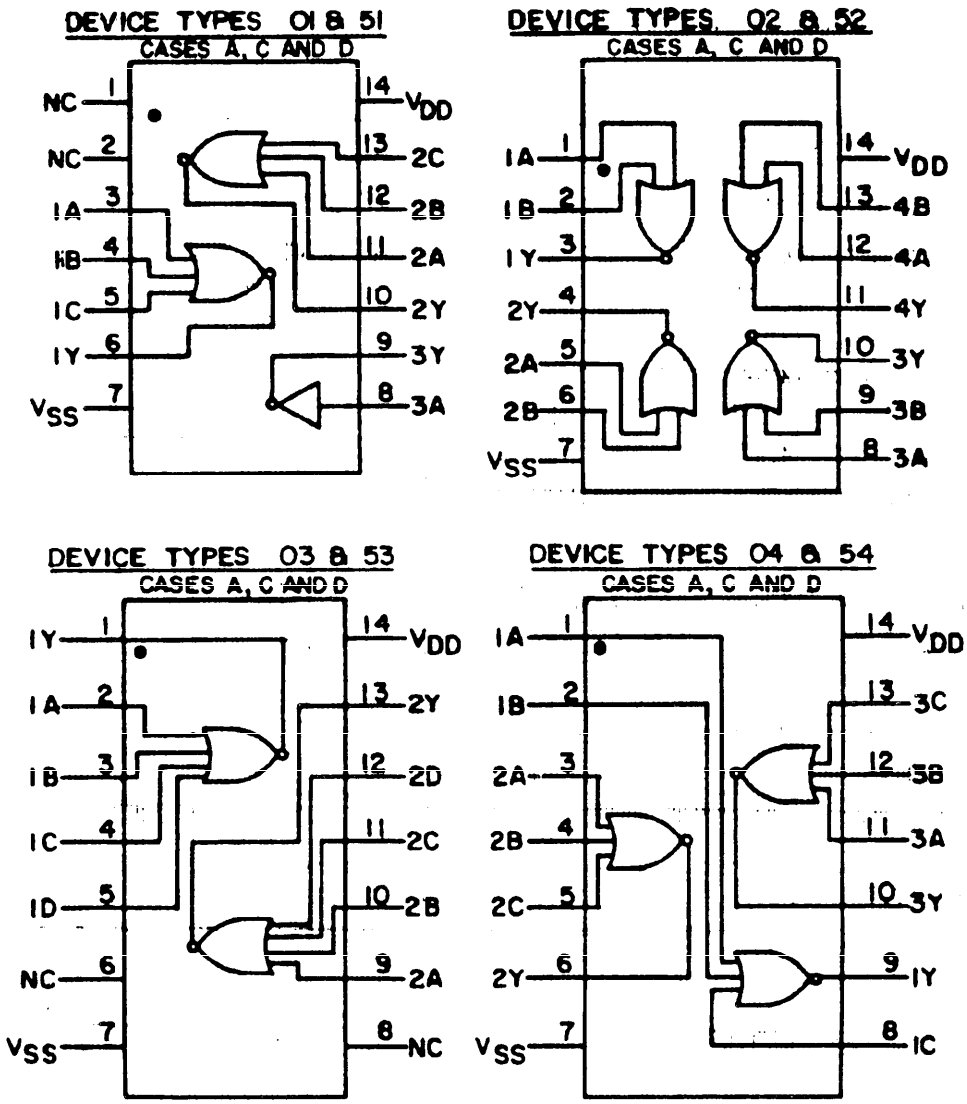


FIGURE 1. Logic diagrams and terminal connections.

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Device types 01, 04, 51, and 54

Truth table each gate			
Input			Output
A	B	C	Y
L	L	L	H
H	L	L	L
L	H	L	L
H	H	L	L
L	L	H	L
H	L	H	L
L	H	H	L
H	H	H	L

Positive logic $Y = \overline{A+B+C}$
(01 Inverter $3Y = \overline{3A}$)

Device types 02 and 52

Truth table each gate		
Input		Output
A	B	Y
L	L	H
H	L	L
L	H	L
H	H	L

Positive logic $Y = \overline{A+B}$

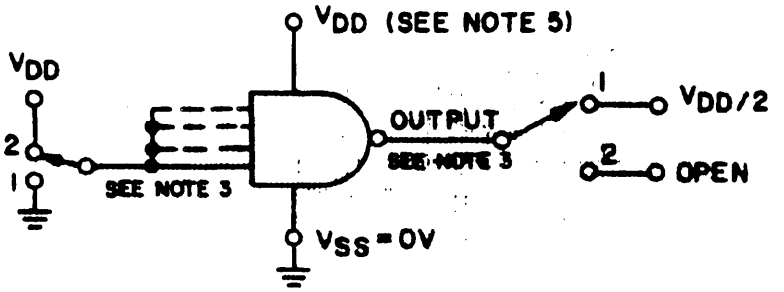
Device types 03 and 53

Truth table each gate				
Input				Output
A	B	C	D	Y
L	L	L	L	H
H	L	L	L	L
L	H	L	L	L
H	H	L	L	L
L	L	H	L	L
H	L	H	L	L
L	H	H	L	L
H	H	H	L	L
L	L	L	H	L
H	L	L	H	L
L	H	L	H	L
H	H	L	H	L
L	L	H	H	L
H	L	H	H	L
L	H	H	H	L
H	H	H	H	L

Positive logic $Y = \overline{A+B+C+D}$

FIGURE 2. Truth tables and logic equations.

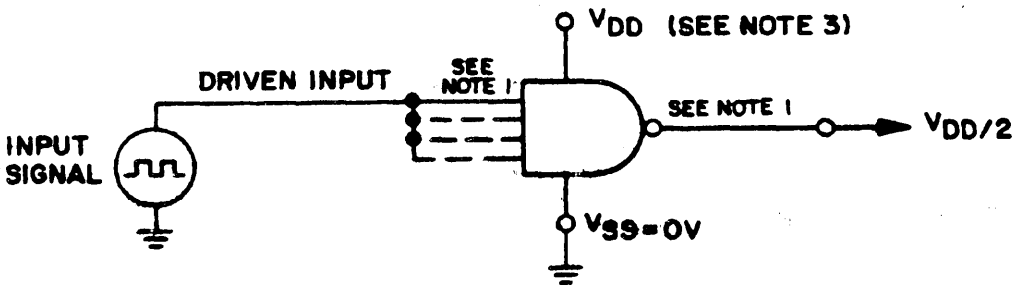
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NOTES:
1. For static burn-in I, all inputs are connected to 0 volts, switch position 1.
2. For static burn-in II, all inputs are connected to V_{DD}, switch position 2.
3. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
4. Output may be in switch position 1 or 2.
5. V_{DD} = 12.5 V minimum, 15.0 V maximum for device types 01, 02, 03, and 04.
   V_{DD} = 15.0 V minimum, 18.0 V maximum for device types 51, 52, 53, and 54.
   $V_{DD/2} = V_{DD/2} \pm 1.0$ V for all devices.
   V_{SS} = 0.0 V.

FIGURE 3. Static burn-in test circuits.

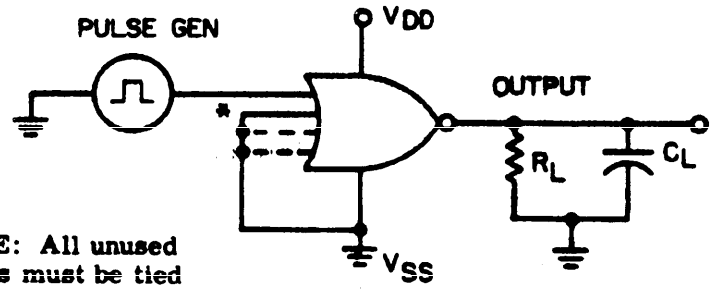
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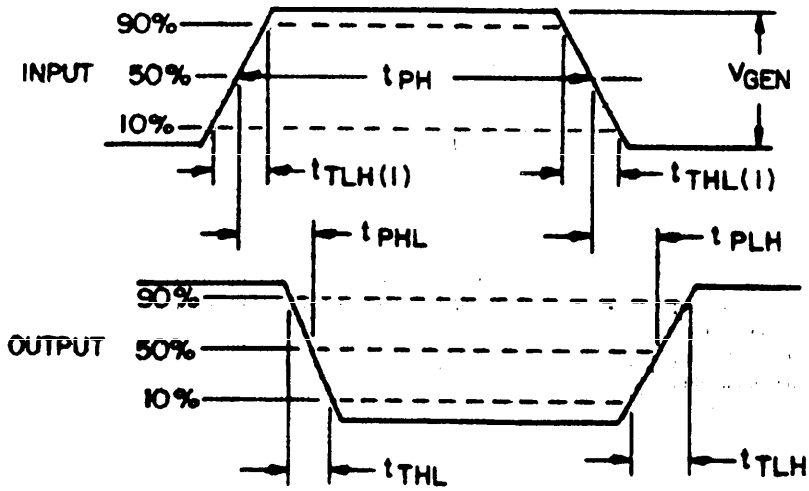


NOTES:
1. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
2. Input signal requirements:
   a. Square wave, 50% duty cycle.
   b. 25 kHz < PRR < 1 MHz.
   c. t_{TLH} and t_{THL} < 1 µs.
   d. Voltage level:
      Minimum = V_{SS} -0.5 V, +10% V_{DD}.
      Maximum = V_{DD} +0.5, -10% V_{DD}.
3. V_{DD} = 12.5 V minimum, 15.0 V maximum for device 01, 02, 03, and 04.
   V_{DD} = 15 V minimum, 18 V maximum for device 51, 52, 53, and 54.
   $V_{DD/2}$ = $V_{DD}/2$ ±1.0 V.
   V_{SS} = 0.0 V.

FIGURE 4. Dynamic burn-in and steady state life test circuit.

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Input pulse

$V_{GEN} = V_{DD} \pm 1.0\%$
$t_{PH} = 1.0 \pm 0.1\ \mu s$
$t_{THL(1)} = t_{TLH(1)} = 10 \pm 2$ ns
Pulse repetition period = $5.0 \pm 1\ \mu s$

FIGURE 5. Switching time test circuit and waveforms.

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TABLE III. Group A insp. for device type 01.

Symbol	MIL-STD-883 method	Cases A, C, D X, Y	TERMINAL CONDITIONS 1/														Meas terminal	Test limits						Units
																		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
			NC	NC	1A	1B	1C	1Y	V_{SS}	3A	3Y	2Y	2A	2B	2C	V_{DD}								
V_{IC} (pos)		1			1 mA											GND	1A		1.5					Vdc
		2				1 mA											1B							
		3					1 mA										1C							
		4								1 mA							3A							
		5											1 mA				2A							
		6												1 mA			2B							
		7													1 mA		2C							
V_{IC} (neg)		8			-1 mA				GND								1A		-6					
		9				-1 mA											1B							
		10					-1 mA										1C							
		11								-1 mA							3A							
		12											-1 mA				2A							
		13												-1 mA			2B							
		14													-1 mA		2C							
I_{SS}	3005 2/	15			15.0 V	GND	GND			15.0 V			15.0 V	GND	GND	15.0 V	V_{SS}		-25		-750			nA
		16			GND	15.0 V	GND			15.0 V			GND	15.0 V	GND									
		17			GND	GND	15.0 V			15.0 V			GND	GND	15.0 V									
		18			GND	GND	GND			GND			GND	GND	GND									
V_{OHA}	3006	19			3/	3/	3/	4/		4.5 V			4.5 V	4.5 V	4.5 V	4.5 V	1Y	2.5		2.5		2.5		Vdc
		20			4.5 V	4.5 V	4.5 V			4.5 V		4/	3/	3/	3/	4.5 V	2Y	2.5		2.5		2.5		
		21			4.5 V	4.5 V	4.5 V			3/	4/		4.5 V	4.5 V	4.5 V	4.5 V	3Y	2.5		2.5		2.5		
V_{OHB}		22			3/	3/	3/	5/		5.0 V			5.0 V	5.0 V	5.0 V	5.0 V	1Y	4.2		4.2		4.2		
		23			5.0 V	5.0 V	5.0 V			5.0 V		5/	3/	3/	3/		2Y	4.2		4.2		4.2		
		24			5.0 V	5.0 V	5.0 V			3/	5/		5.0 V	5.0 V	5.0 V		3Y	4.2		4.2		4.2		
V_{OHC}		25			3/	3/	3/			5.0 V			5.0 V	5.0 V	5.0 V		1Y	4.95		4.95		4.95		
		26			5.0 V	5.0 V	5.0 V			5.0 V			3/	3/	3/		2Y	4.95		4.95		4.95		
		27			5.0 V	5.0 V	5.0 V			3/			5.0 V	5.0 V	5.0 V		3Y	4.95		4.95		4.95		
V_{OH4}		28			6/	6/	6/			12.5 V			12.5 V	12.5 V	12.5 V	12.5 V	1Y	11.25		11.25		11.25		
		29			12.5 V	12.5 V	12.5 V			12.5 V			6/	6/	6/	12.5 V	2Y	11.25		11.25		11.25		
		30			12.5 V	12.5 V	12.5 V			6/			12.5 V	12.5 V	12.5 V	12.5 V	3Y	11.25		11.25		11.25		

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	TERMINAL CONDITIONS 1/														Meas terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test No.	NC	NC	1A	1B	1C	1Y	V_{SS}	3A	3Y	2Y	2A	2B	2C	V_{PO}		Min	Max	Min	Max	Min	Max	
V_{OL1}	3007	31			7/	GND	GND	8/	GND	GND			GND	GND	GND	5.5 V	1Y		0.40		0.5		0.40	Vdc
		32			GND	7/	GND	8/					GND				1Y							
		33				GND	7/	8/					GND	7/	7/		1Y							
		34					GND					8/	7/	GND			2Y							
		35										8/	GND	7/	7/		2Y							
		36										8/	7/	GND			2Y							
		37								7/	8/				GND		3Y							
V_{OL2}		38			7/	GND	GND	15/	GND	GND			GND	GND	GND	3.0 V	1Y		0.7		0.7		0.7	
		39			GND	7/	GND	15/					GND				1Y							
		40				GND	7/	15/					GND	7/	7/		1Y							
		41					GND					15/	7/	GND			2Y							
		42										15/	GND	7/	7/		2Y							
		43										15/	7/	GND			2Y							
		44								7/	15/				GND		3Y							
V_{OL3}		45			7/	GND	GND	15/	GND	GND			GND	GND	GND	3.0 V	1Y		50		50		50	mV
		46			GND	7/	GND	15/					GND				1Y							
		47				GND	7/	15/					GND	7/	7/		1Y							
		48					GND					15/	7/	GND			2Y							
		49										15/	GND	7/	7/		2Y							
		50										15/	7/	GND			2Y							
		51								7/	15/				GND		3Y							
V_{OL4}		52			9/	GND	GND	16/	GND	GND			GND	GND	GND	12.5 V	1Y		1.25		1.25		1.25	Vdc
		53			GND	9/	GND	16/					GND				1Y							
		54				GND	9/	16/					GND	9/	9/		1Y							
		55					GND					16/	9/	GND			2Y							
		56										16/	GND	9/	9/		2Y							
		57										16/	9/	GND			2Y							
		58								9/	16/				GND		3Y							
V_{OL5}		59			7/	7/	7/	17/				17/			GND	5.0 V	1Y		0.40		0.5		0.40	
		60			GND	GND	GND	17/				17/	7/	7/	7/	5.0 V	2Y		0.40		0.5		0.40	

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-883 method	Cases A, C, D X, Y Test No.	TERMINAL CONDITIONS 1/ 1 NC	2 NC	3 1A	4 1B	5 1C	6 1Y	7 V_{ss}	8 3A	9 3Y	10 2Y	11 2A	12 2B	13 2C	14 V_{DD}	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Subgroup 1 T_A = 25°C Max	Subgroup 2 T_A = 125°C Min	Subgroup 2 T_A = 125°C Max	Subgroup 3 T_A = -55°C Min	Subgroup 3 T_A = -55°C Max	Units
I_{IL} 11/	3009	61			GND	GND	GND		GND	GND			GND	GND	GND	15.0 V	All inputs together		-7.0					nA
		62															1A		-1.0		-45			
		63															1B							
		64															1C							
		65															2A							
		66															2B							
		67															2C							
		68															3A							
I_{IH} 11/	3010	69			15.0 V	15.0 V	15.0 V			15.0 V			15.0 V	15.0 V	15.0 V		All inputs together		7.0					
		70			15.0 V	GND	GND			GND			GND	GND	GND		1A		1.0		45			
		71			GND	15.0 V	GND						GND				1B							
		72				GND	15.0 V						GND				1C							
		73					GND						15.0 V				2A							
		74											GND	15.0 V			2B							
		75											GND	GND	15.0 V		2C							
		76								15.0 V			GND	GND	GND		3A							
																		Subgroup 4 T_A = 25°C						
C_I	3012	77			12/				GND							GND	1A		12					pF
		78				12/											1B							
		79					12/										1C							
		80											12/				2A							
		81												12/			2B							
		82													12/		2C							
		83								12/							3A							

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y Test No.	TERMINAL CONDITIONS 1/ 1 NC	2 NC	3 1A	4 1B	5 1C	6 1Y	7 V_{ss}	8 3A	9 3Y	10 2Y	11 2A	12 2B	13 2C	14 V_{DD}	Meas terminal	Test limits Subgroup 9 T_A = 25°C Min	Subgroup 9 T_A = 25°C Max	Subgroup 10 T_A = 125°C Min	Subgroup 10 T_A = 125°C Max	Subgroup 11 T_A = -55°C Min	Subgroup 11 T_A = -55°C Max	Units
t_{PHL}	3003	84			13/	GND	GND	14/	GND	GND			GND	GND	GND	5.0 V	1A to 1Y	11	210	16	315	11	210	ns
	Fig. 5	85			GND	13/	GND	14/					GND				1B to 1Y							
		86				GND	13/	14/					GND				1C to 1Y							
		87					GND					13/	13/				2A to 2Y							
		88										13/	GND	13/			2B to 2Y							
		89										13/		GND	13/		2C to 2Y							
		90								13/	14/				GND		3A to 3Y							
t_{PLH}		91			13/			14/		GND							1A to 1Y	14	275	21	415	14	275	
		92			GND	13/		14/									1B to 1Y							
		93				GND	13/	14/									1C to 1Y							
		94					GND					14/	13/				2A to 2Y							
		95										14/	GND	13/			2B to 2Y							
		96										14/		GND	13/		2C to 2Y							
		97								13/	14/			GND	GND		3A to 3Y							
t_{THL}	3004	98			13/	GND	GND	14/		GND				GND	GND		1Y	15	300	23	450	15	300	ns
	Fig. 5	99			GND	13/	GND	14/					GND				1Y							
		100				GND	13/	14/					GND				1Y							
		101					GND					14/	13/				2Y							
		102										14/	GND	13/			2Y							
		103										14/		GND	13/		2Y							
		104								13/	14/				GND		3Y							
t_{TLH}		105			13/			14/		GND							1Y	23	450	34	675	23	450	
		106			GND	13/		14/									1Y							
		107				GND	13/	14/									1Y							
		108					GND					14/	13/				2Y							
		109										14/	GND	13/			2Y							
		110										14/		GND	13/		2Y							
		111								13/	14/			GND	GND		3Y							

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TABLE III. Group A [illegible]tion for device type 02.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	TERMINAL CONDITIONS 1/															Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test No.	1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1	1 mA													GND	1A		1.5					Vdc
		2		1 mA													1B							
		3					1 mA										2A							
		4						1 mA									2B							
		5								1 mA							3A							
		6									1 mA						3B							
		7												1 mA			4A							
		8													1 mA		4B							
V_{IC} (neg)		9	-1 mA						GND								1A		-6					
		10		-1 mA													1B							
		11					-1 mA										2A							
		12						-1 mA									2B							
		13								-1 mA							3A							
		14									-1 mA						3B							
		15												-1 mA			4A							
		16													-1 mA		4B							
I_{ss}	3005	17	15.0 V	GND			GND	15.0 V		15.0 V	GND			GND	15.0 V	15.0 V	V_{ss}		-25.0		-750			nA
	2/	18	GND	15.0 V			15.0 V	GND		GND	15.0 V			15.0 V	GND	15.0 V	V_{dd}		-25.0		-750			nA
		19	GND	GND			GND	GND		GND	GND			GND	GND	15.0 V	V_{ss}		-25.0		-750			nA
V_{OH1}	3006	20	3/	3/	4/		4.5 V	4.5 V		4.5 V	4.5 V			4.5 V	4.5 V	4.5 V	1Y	2.5		2.5		2.5		Vdc
		21	4.5 V	4.5 V		4/	3/	3/		4.5 V	4.5 V			4.5 V	4.5 V		2Y							
		22					4.5 V	4.5 V		3/	3/	4/		4.5 V	4.5 V		3Y							
		23					4.5 V	4.5 V		4.5 V	4.5 V		4/	3/	3/		4Y							
V_{OH2}		24	3/	3/	5/		5.0 V	5.0 V		5.0 V	5.0 V			5.0 V	5.0 V	5.0 V	1Y	4.2		4.2		4.2		
		25	5.0 V	5.0 V		5/	3/	3/		5.0 V	5.0 V			5.0 V	5.0 V		2Y							
		26	5.0 V	5.0 V			5.0 V	5.0 V		3/	3/	5/		5.0 V	5.0 V		3Y							
		27	5.0 V	5.0 V			5.0 V	5.0 V		5.0 V	5.0 V		5/	3/	3/		4Y							
V_{OH3}		28	3/	3/			5.0 V	5.0 V		5.0 V	5.0 V			5.0 V	5.0 V		1Y	4.95		4.95		4.95		
		29	5.0 V	5.0 V			3/	3/		5.0 V	5.0 V			5.0 V	5.0 V		2Y							
		30	5.0 V	5.0 V			5.0 V	5.0 V		3/	3/			5.0 V	5.0 V		3Y							

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	TERMINAL CONDITIONS 1/														Meas terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = −55°C		
		Test No.	1A	1B	1Y	2Y	2A	2B	V_{NR}	3A	3B	3Y	4Y	4A	4B	V_{PD}		Min	Max	Min	Max	Min	Max	
V_{OH3}	3006	31	5.0 V	5.0 V			5.0 V	5.0 V	GND	5.0 V	5.0 V			3/	3/	5.0 V	4Y	4.95		4.95		4.95		Vdc
V_{OH4}		32	4/	4/			12.5 V	12.5 V		12.5 V	12.5 V			12.5 V	12.5 V	12.5 V	1Y	11.25		11.25		11.25		
		33	12.5 V	12.5 V			4/	4/		12.5 V	12.5 V			12.5 V	12.5 V		2Y							
		34	12.5 V	12.5 V			12.5 V	12.5 V		4/	4/			12.5 V	12.5 V		3Y							
		35	12.5 V	12.5 V			12.5 V	12.5 V		12.5 V	12.5 V			4/	4/		4Y							
V_{OL1}	3007	36	1/	GND	8/		GND	GND		GND	GND			GND	GND	5.5 V	1Y		0.40		0.50		0.40	
		37	GND	1/	8/		GND	GND									1Y							
		38		GND		8/	1/	GND									2Y							
		39				8/	GND	1/									2Y							
		40						GND		1/		8/					3Y							
		41								GND	1/	8/					3Y							
		42									GND		8/	1/			4Y							
		43											8/	GND	1/		4Y							
V_{OL2}		44	1/		15/			GND							GND	5.0 V	1Y		0.70		0.70		0.70	
		45	GND	1/	15/			GND									1Y							
		46		GND		15/	1/	GND									2Y							
		47				15/	GND	1/									2Y							
		48						GND		1/		15/					3Y							
		49								GND	1/	15/					3Y							
		50								GND	GND		15/	1/			4Y							
		51								GND	GND		15/	GND	1/		4Y							
V_{OL3}		52	1/	GND			GND	GND	GND	GND	GND				GND		1Y		50		50		50	mV
		53	GND	1/			GND	GND									1Y							
		54		GND			1/	GND									2Y							
		55					GND	1/									2Y							
		56						GND		1/							3Y							
		57								GND	1/						3Y							
		58									GND			1/			4Y							
		59												GND	1/		4Y							

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Cases A, C, D X, Y	TERMINAL CONDITIONS 1/														Meas terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = −55°C		
		Test No.	1A	1B	1Y	2Y	2A	1Y	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{OL3}	3007	60	9/	GND			GND	GND	GND	GND	GND			GND	GND	12.5 V	1Y		1.25		1.25		1.25	Vdc
		61	GND	9/			GND	GND									1Y							
		62		GND			9/	GND									2Y							
		63					GND	9/									2Y							
		64						GND		9/							3Y							
		65								GND	9/						3Y							
		66									GND			9/			4Y							
		67												GND	9/		4Y							
V_{OL5}		68	7/	7/	10/										GND	5.0 V	1Y		0.40		0.5		0.40	
		69	GND	GND		10/	7/	7/							GND		2Y							
		70					GND	GND		7/	7/	10/			GND		3Y							
		71								GND	GND		10/	7/	7/		4Y							
I_{IL}	3009 11/	72												GND	GND	15.0 V	All inputs together		-8.0					mA
		73															1A		-1.0		-45			
		74															1B							
		75															2A							
		76															2B							
		77															3A							
		78															3B							
		79															4A							
		80															4B							
I_{IH}	3010 11/	81	15.0 V	15.0 V			15.0 V	15.0V		15.0 V	15.0 V			15.0 V	15.0 V		All inputs together		8.0					
		82	15.0 V	GND			GND	GND		GND	GND			GND	GND		1A		1.0		45			
		83	GND	15.0 V			GND	GND									1B							
		84		GND			15.0 V	GND									2A							
		85		GND			GND	15.0 V									2B							

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D X, Y Test No.	TERMINAL CONDITIONS 1/ 1 1A	2 1B	3 1Y	4 2Y	5 2A	6 2B	7 V_{CC}	8 3A	9 3B	10 3Y	11 4Y	12 4A	13 4B	14 V_{IH1}	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Max	Subgroup 2 T_A = 125°C Min	Max	Subgroup 3 T_A = -55°C Min	Max	Units
I_{IH}	3010 11/	86	GND	GND			GND	GND	GND	15.0 V	GND			GND	GND	15.0 V	3A		1.0		45			nA
		87								GND	15.0 V			GND			3B							
		88								GND	GND			15.0 V			4A							
		89								GND	GND			GND	15.0 V		4B							
																		Subgroup 4 T_A = 25°C						
C_I	3012	90	12/						GND							GND	1A		12					pF
		91		12/													1B							
		92					12/										2A							
		93						12/									2B							
		94								12/							3A							
		95									12/						3B							
		96												12/			4A							
		97													12/		4B							
																		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
t_{PHL}	3003 Fig. 5	98	13/	GND	14/		GND	GND	GND	GND	GND			GND	GND	5.0 V	1A to 1Y	30	210	39	315	15	210	ns
		99	GND	13/	14/		GND	GND									1B to 1Y							
		100		GND		14/	13/	GND									2A to 2Y							
		101				14/	GND	13/									2B to 2Y							
		102						GND		13/		14/					3A to 3Y							
		103								GND	13/	14/					3B to 3Y							
		104									GND		14/	13/			4A to 4Y							
		105											14/	GND	13/		4B to 4Y							
t_{PLH}		106	13/		14/										GND		1A to 1Y					22		
		107	GND	13/	14/												1B to 1Y							
		108		GND		14/	13/										2A to 2Y							
		109				14/	GND	13/									2B to 2Y							
		110						GND		13/		14/					3A to 3Y							
		111								GND	13/	14/					3B to 3Y							
		112									GND		14/	13/			4A to 4Y							
		113									GND		14/	GND	13/		4B to 4Y							

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y Test No.	TERMINAL CONDITIONS 1/ 1 1A	2 1B	3 1Y	4 2Y	5 2A	6 2B	7 V_{ss}	8 3A	9 3B	10 3Y	11 4Y	12 4A	13 4B	14 V_{cc}	Meas terminal	Test limits Subgroup 9 T_A = 25°C Min	Subgroup 9 T_A = 25°C Max	Subgroup 10 T_A = 125°C Min	Subgroup 10 T_A = 125°C Max	Subgroup 11 T_A = −55°C Min	Subgroup 11 T_A = −55°C Max	Units
t_{THL}	3004	114	13/	GND	14/		GND	GND	GND	GND	GND			GND	GND	5.0 V	1Y	40	300	52	450	30	300	ns
	Fig. 5	115	GND	13/	14/		GND										1Y							
		116		GND		14/	13/										2Y							
		117				14/	GND	13/									2Y							
		118						GND		13/		14/					3Y							
		119								GND	13/	14/					3Y							
		120									GND		14/	13/			4Y							
		121											14/	GND	13/		4Y							
t_{TLH}		122	13/		14/										GND		1Y	50	410	65	615	37	410	
		123	GND	13/	14/												1Y							
		124		GND		14/	13/										2Y							
		125				14/	GND	13/									2Y							
		126						GND		13/		14/					3Y							
		127								GND	13/	14/					3Y							
		128								GND	GND		14/	13/			4Y							
		129								GND	GND		14/	GND	13/		4Y							

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TABLE III. Group A inspection for device type 03.

Symbol	MIL-STD-883 method	Cases A, C, D X, Y Test No.	TERMINAL CONDITIONS 1/ 1 1Y	2 1A	3 1B	4 1C	5 1D	6 NC	7 V_{EE}	8 NC	9 2A	10 2B	11 2C	12 2D	13 2Y	14 V_{CC}	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Max	Subgroup 2 T_A = 125°C Min	Max	Subgroup 3 T_A = -55°C Min	Max	Units
V_{IC} (pos)		1		1 mA												GND	1A		1.5					Vdc
		2			1 mA												1B							
		3				1 mA											1C							
		4					1 mA										1D							
		5									1 mA						2A							
		6										1 mA					2B							
		7											1 mA				2C							
		8												1 mA			2D							
V_{IC} (neg)		9		-1 mA					GND								1A		-6					
		10			-1 mA												1B							
		11				-1 mA											1C							
		12					-1 mA										1D							
		13									-1 mA						2A							
		14										-1 mA					2B							
		15											-1 mA				2C							
		16												-1 mA			2D							
I_{IH}	3086 2/	17		GND	GND	GND	GND				GND	GND	GND	GND		15.0 V	V_{IH}		-25		-750			nA
		18			GND	GND	15.0 V				15.0 V	GND	GND											
		19			GND	15.0 V	GND				GND	15.0 V	GND											
		20			15.0 V	GND	GND				GND	GND	15.0 V											
		21		15.0 V	GND	GND	GND				GND	GND	GND	15.0 V										
V_{OH1}	3006	22	4/	3/	3/	3/	3/				4.5 V	4.5 V	4.5 V	4.5 V		4.5 V	1Y	2.5		2.5		2.5		Vdc
		23		4.5 V	4.5 V	4.5 V	4.5 V				3/	3/	3/	3/	4/	4.5 V	2Y	2.5		2.5		2.5		
V_{OH2}		24	5/	3/	3/	3/	3/				5.0 V	5.0 V	5.0 V	5.0 V		5.0 V	1Y	4.2		4.2		4.2		
		25		5.0 V	5.0 V	5.0 V	5.0 V				3/	3/	3/	3/	5/		2Y	4.2		4.2		4.2		
V_{OH3}		26		3/	3/	3/	3/				5.0 V	5.0 V	5.0 V	5.0 V			1Y	4.95		4.95		4.95		
		27		5.0 V	5.0 V	5.0 V	5.0 V				3/	3/	3/	3/			2Y	4.95		4.95		4.95		
V_{OH4}		28		6/	6/	6/	6/				12.5 V	12.5 V	12.5 V	12.5 V		12.5 V	1Y	11.25		11.25		11.25		
		29		12.5 V	12.5 V	12.5 V	12.5 V				6/	6/	6/	6/		12.5 V	2Y	11.25		11.25		11.25		

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TABLE III. Group A ion for device type 03 – Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y Test No.	TERMINAL CONDITIONS 1/ 1 1Y	2 1A	3 1B	4 1C	5 1D	6 NC	7 V_{SS}	8 NC	9 2A	10 2B	11 2C	12 2D	13 2Y	14 V_{DD}	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Subgroup 1 T_A = 25°C Max	Subgroup 2 T_A = 125°C Min	Subgroup 2 T_A = 125°C Max	Subgroup 3 T_A = −55°C Min	Subgroup 3 T_A = −55°C Max	Units
V_{OL1}	3007	30	8/	1/	GND	GND	GND		GND		GND	GND	GND	GND		5.5 V	1Y		0.40		0.5		0.40	Vdc
		31		GND	1/	GND	GND										1Y							
		32			GND	1/	GND										1Y							
		33				GND	1/										1Y							
		34					GND				1/				8/		2Y							
		35									GND	1/					2Y							
		36										GND	1/				2Y							
		37											GND	1/			2Y							
V_{OL2}		38	15/	1/										GND		5.0 V	1Y		0.7		0.7		0.7	
		39		GND	1/												1Y							
		40			GND	1/											1Y							
		41				GND	1/										1Y							
		42					GND				1/				15/		2Y							
		43									GND	1/					2Y							
		44										GND	1/				2Y							
		45											GND	1/			2Y							
V_{OL3}		46		1/										GND			1Y		50		50		50	mV
		47		GND	1/												1Y							
		48			GND	1/											1Y							
		49				GND	1/										1Y							
		50					GND				1/						2Y							
		51									GND	1/					2Y							
		52										GND	1/				2Y							
		53											GND	1/			2Y							
V_{OL4}		54		9/										GND		12.5 V	1Y		1.25		1.25		1.25	Vdc
		55		GND	9/												1Y							
		56			GND	9/											1Y							
		57				GND	9/										1Y							
		58					GND				9/						2Y							
		59									GND	9/					2Y							
		60									GND	GND	9/				2Y							
		61									GND	GND	GND	9/			2Y							

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D X, Y Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Test limits: Subgroup 1 T_A = 25°C Min	Subgroup 1 T_A = 25°C Max	Subgroup 2 T_A = 125°C Min	Subgroup 2 T_A = 125°C Max	Subgroup 3 T_A = -55°C Min	Subgroup 3 T_A = -55°C Max	Units
V_{OL3}	3007	62	10/	1/	1/	1/	1/		GND		GND	GND	GND	GND	2Y	V_{DD}	1Y		0.40		0.5		0.40	Vdc
		63		GND	GND	GND	GND				1/	1/	1/	1/	10/	5.0 V	2Y		0.40		0.5		0.40	Vdc
I_{IL} 11/	3009	64														15.0 V	All inputs together		-8.0					nA
		65															1A		-1.0		-45			
		66															1B							
		67															1C							
		68															1D							
		69															2A							
		70															2B							
		71															2C							
		72									1/	1/	1/	1/			2D							
I_{IH} 11/	3010	73		15.0 V	15.0 V	15.0 V	15.0 V				15.0 V	15.0 V	15.0 V	15.0 V			All inputs together		8.0					nA
		74		15.0 V	GND	GND	GND				GND	GND	GND	GND			1A		1.0		45			
		75		GND	15.0 V	GND	GND				GND	GND					1B							
		76			GND	15.0 V	GND				GND						1C							
		77				GND	15.0 V				GND						1D							
		78					GND				15.0 V						2A							
		79									GND	15.0 V					2B							
		80									GND	GND	15.0 V				2C							
		81									GND	GND	GND	15.0 V			2D							
C_I	3012	82		12/					GND							GND	1A	Subgroup 4 T_A = 25°C	12					pF
		83			12/												1B							
		84				12/											1C							
		85					12/										1D							
		86									12/						2A							
		87										12/					2B							
		88											12/				2C							
		89												12/			2D							

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	TERMINAL CONDITIONS 1/															Meas terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14			Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test No.	1Y	1A	1B	1C	1D	NC	V_{SS}	NC	2A	2B	2C	2D	2Y	V_{DD}			Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 Fig. 5	90	14/	13/	GND	GND	GND		GND		GND	GND	GND	GND		5.0 V	1A to 1Y		11	210	16	315	11	210	ns
		91		GND	13/	GND	GND										1B to 1Y								
		92			GND	13/	GND										1C to 1Y								
		93				GND	13/										1D to 1Y								
		94					GND				13/				14/		2A to 2Y								
		95									GND	13/					2B to 2Y								
		96										GND	13/				2C to 2Y								
		97											GND	13/			2D to 2Y								
t_{PLH}		98	14/	13/										GND			1A to 1Y								
		99		GND	13/												1B to 1Y								
		100			GND	13/											1C to 1Y								
		101				GND	13/										1D to 1Y								
		102					GND				13/				14/		2A to 2Y								
		103									GND	13/					2B to 2Y								
		104										GND	13/				2C to 2Y								
		105											GND	13/			2D to 2Y								
t_{THL}	3004 Fig. 5	106	14/	13/										GND			1Y		15	300	23	450	15	300	
		107		GND	13/												1Y								
		108			GND	13/											1Y								
		109				GND	13/										1Y								
		110					GND				13/				14/		2Y								
		111									GND	13/					2Y								
		112										GND	13/				2Y								
		113											GND	13/			2Y								
t_{TLH}		114	14/	13/										GND			1Y		21	410	31	615	21	410	
		115		GND	13/												1Y								
		116			GND	13/											1Y								
		117				GND	13/										1Y								
		118					GND				13/				14/		2Y								
		119									GND	13/					2Y								
		120									GND	GND	13/				2Y								
		121									GND	GND	GND	13/			2Y								

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TABLE III. Group A inspection for device type 04.

Symbol	MIL-STD-883 method	Cases A, C, D X, Y	TERMINAL CONDITIONS 1/															Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = −55°C		
		Test No.	1A	1B	2A	2B	2C	2Y	V_{SS}	1C	1Y	3Y	3A	3B	3C	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{IK} (pos)		1	1 mA													GND	1A		1.5					Vdc
		2		1 mA													1B							
		3								1 mA							1C							
		4			1 mA												2A							
		5				1 mA											2B							
		6					1 mA										2C							
		7											1 mA				3A							
		8												1 mA			3B							
		9													1 mA		3C							
V_{IK} (neg)		10	−1 mA						GND								1A		−6					
		11		−1 mA													1B							
		12								−1 mA							1C							
		13			−1 mA												2A							
		14				−1 mA											2B							
		15					−1 mA										2C							
		16											−1 mA				3A							
		17												−1 mA			3B							
		18													−1 mA		3C							
I_{SS}	3005 2/	19	15.0 V	GND	15.0 V	GND	GND			GND			15.0 V	GND	GND	15.0 V	V_{SS}		−25.0		−750			nA
		20	GND	15.0 V	GND	15.0 V	GND			GND			GND	15.0 V	GND									
		21	GND	GND	GND	GND	15.0 V			15.0 V			GND	GND	15.0 V									
		22	GND	GND	GND	GND	GND			GND			GND	GND	GND									
V_{OH1}	3006	23	3/	3/	4.5 V	4.5 V	4.5 V			3/	4/		4.5 V	4.5 V	4.5 V	4.5 V	1Y	2.5		2.5		2.5		Vdc
		24	4.5 V	4.5 V	3/	3/	3/	4/		4.5 V			4.5 V	4.5 V	4.5 V	4.5 V	2Y	2.5		2.5		2.5		
		25	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V			4.5 V		4/	3/	3/	3/	4.5 V	3Y	2.5		2.5		2.5		
V_{OH2}		26	3/	3/	5.0 V	5.0 V	5.0 V			3/	5/		5.0 V	5.0 V	5.0 V	5.0 V	1Y	4.2		4.2		4.2		
		27	5.0 V	5.0 V	3/	3/	3/	5/		5.0 V			5.0 V	5.0 V	5.0 V		2Y	4.2		4.2		4.2		
		28	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V			5.0 V		5/	3/	3/	3/		3Y	4.2		4.2		4.2		
V_{OH3}		29	3/	3/	5.0 V	5.0 V	5.0 V			3/			5.0 V	5.0 V	5.0 V		1Y	4.95		4.95		4.95		
		30	5.0 V	5.0 V	3/	3/	3/			5.0 V			5.0 V	5.0 V	5.0 V		2Y	4.95		4.95		4.95		
		31	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V			5.0 V			3/	3/	3/		3Y	4.95		4.95		4.95		

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TABLE III. Group A ... dica for device type 04 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y Test No.	TERMINAL CONDITIONS 1/ 1 1A	2 1B	3 2A	4 2B	5 2C	6 2Y	7 V_{EE}	8 1C	9 1Y	10 3Y	11 3A	12 3B	13 3C	14 V_{CC}	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Subgroup 1 T_A = 25°C Max	Subgroup 2 T_A = 125°C Min	Subgroup 2 T_A = 125°C Max	Subgroup 3 T_A = -55°C Min	Subgroup 3 T_A = -55°C Max	Units
V_{OH1}	3006	32	6/	6/	12.5 V	12.5 V	12.5 V		GND	6/			12.5 V	12.5 V	12.5 V	12.5 V	1Y	11.25		11.25		11.25		Vdc
		33	12.5 V	12.5 V	6/	6/	6/			12.5 V			12.5 V	12.5 V	12.5 V		2Y	11.25		11.25		11.25		
		34	12.5 V	12.5 V	12.5 V	12.5 V	12.5 V			12.5 V			6/	6/	6/		3Y	11.25		11.25				
V_{OL1}	3007	35	1/	GND	GND	GND	GND			GND	6/		GND	GND	GND	5.5 V	1Y		0.40		0.5		0.40	
		36	GND	1/	GND	GND				GND	6/		GND	GND	1/		1Y							
		37	GND	GND	1/	1/				1/	6/		1/	1/			1Y							
		38	1/	1/				6/		GND	6/						2Y							
		39	GND	GND				6/			6/						2Y							
		40	1/	1/				6/			6/						2Y							
		41	GND	GND								6/	1/	1/			3Y							
		42	1/	1/								6/	GND	1/			3Y							
		43	GND	GND								6/	GND	GND			3Y							
V_{OL2}	3007	44	1/	GND	GND	GND	GND			GND	1/		GND	GND	GND	5.0 V	1Y		0.7		0.7		0.7	
		45	GND	1/	GND	GND				GND	1/		GND	GND	1/		1Y							
		46	GND	GND	1/	1/				1/	1/		1/	1/			1Y							
		47	1/	1/				1/		GND	1/						2Y							
		48	GND	GND				1/			1/						2Y							
		49	1/	1/				1/			1/						2Y							
		50	GND	GND								1/	1/	1/			3Y							
		51	1/	1/								1/	GND	1/			3Y							
		52	GND	GND								1/	GND	GND			3Y							
V_{OL3}	3007	53	1/	GND	GND	GND	GND			GND	1/		GND	GND	GND	4.5 V	1Y		50		50		50	mV
		54	GND	1/	GND	GND				GND	1/		GND	GND	1/		1Y							
		55	GND	GND	1/	1/				1/	1/		1/	1/			1Y							
		56	1/	1/				1/		GND	1/						2Y							
		57	GND	GND				1/			1/						2Y							
		58	1/	1/				1/			1/						2Y							
		59	GND	GND								1/	1/	1/			3Y							
		60	1/	1/								1/	GND	1/			3Y							
		61	GND	GND								1/	GND	GND			3Y							

See notes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y Test No.	TERMINAL CONDITIONS 1/ 1 1A	2 1B	3 2A	4 2B	5 2C	6 2Y	7 V_{AA}	8 1C	9 1Y	10 3Y	11 3A	12 3B	13 3C	14 V_{DD}	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Subgroup 1 T_A = 25°C Max	Subgroup 2 T_A = 125°C Min	Subgroup 2 T_A = 125°C Max	Subgroup 3 T_A = -55°C Min	Subgroup 3 T_A = -55°C Max	Units
V_{OL1}	3007	62	9/	GND	GND	GND	GND		GND	GND			GND	GND	GND	12.5 V	1Y		1.25		1.25		1.25	Vdc
		63	GND	9/													1Y							
		64		GND						9/							1Y							
		65			9/					GND							2Y							
		66			GND	9/											2Y							
		67				GND	9/										2Y							
		68					GND						9/				3Y							
		69											GND	9/			3Y							
		70												GND	9/		3Y							
V_{OL2}		71	7/	7/	GND	GND	GND			7/	10/			GND	GND	5.0 V	1Y		0.4		0.5		0.4	
		72	GND	GND	7/	7/	7/	10/		GND				GND	GND		2Y							
		73	GND	GND	GND	GND	GND					10/	7/	7/	7/		3Y							
I_{IL} 11/	3009	74											GND	GND	GND	15.0 V	All inputs together		-9.0					nA
		75															1A		-1.0		-45			
		76															1B							
		77															1C							
		78															2A							
		79															2B							
		80															2C							
		81															3A							
		82															3B							
		83															3C							
I_{IH} 11/	3010	84	15.0 V	15.0 V	15.0 V	15.0 V	15.0 V			15.0 V			15.0 V	15.0 V	15.0 V	15.0 V	All inputs together		9.0					
		85	15.0 V	GND	GND	GND	GND			GND			GND	GND	GND		1A		1.0		45			
		86	GND	15.0 V	GND					GND							1B							
		87		GND	GND					15.0 V							1C							
		88			15.0 V					GND							2A							
		89			GND	15.0 V											2B							
		90				GND	15.0 V										2C							
		91					GND						15.0 V				3A							
		92					GND						GND	15.0 V			3B							
		93					GND						GND	GND	15.0 V		3C							

See notes at end of device type 04.

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TABLE III. Group A ... ctions for device type 04 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y Test No.	TERMINAL CONDITIONS 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Test limits Subgroup 4 T_A = 25°C Min	Max	Subgroup 9 T_A = 25°C Min	Max	Subgroup 10 T_A = 125°C Min	Max	Subgroup 11 T_A = -55°C Min	Max	Units
C_I	3012	94	1A 12/						GND							V_{DD}	1A		12							pF
		95		1B	12/												1B									
		96				2B	12/										1C									
		97			2A 12/												2A									
		98						2Y					12/				2B									
		99					2C 12/										2C									
		100											3A 12/				3A									
		101												3B 12/			3B									
		102													3C 12/		3C									
I_{DDH}	3012 Fig. 5	103	1A 13/	GND 13/	GND	GND	GND	GND	GND	GND	14/	14/	GND	GND	GND	5.0 V	1A to 1Y		11		210		16		315	mA
		104	GND		GND 13/					GND	14/	14/		GND			1B to 1Y									
		105				GND 13/				13/	14/	14/					1C to 1Y									
		106			GND 13/			14/		GND							2A to 2Y									
		107				GND 13/		14/			14/						2B to 2Y									
		108					GND 13/	14/				14/					2C to 2Y									
		109					GND					14/	GND 13/				3A to 3Y									
		110										14/	GND	13/			3B to 3Y									
		111										14/		GND	13/		3C to 3Y									
I_{DDL}	3003 Fig. 5	112	1A 13/	GND 13/	GND	GND	GND	GND	GND	GND	14/	14/	GND	GND	GND	5.0 V	1A to 1Y		11		210		16		315	mA
		113	GND		GND 13/					GND	14/	14/		GND			1B to 1Y									
		114				GND 13/				13/	14/	14/					1C to 1Y									
		115			GND 13/			14/		GND							2A to 2Y									
		116				GND 13/		14/			14/						2B to 2Y									
		117					GND 13/	14/				14/					2C to 2Y									
		118					GND					14/	GND 13/				3A to 3Y									
		119										14/	GND	13/			3B to 3Y									
		120										14/		GND	13/		3C to 3Y									

See notes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y Test No.	TERMINAL CONDITIONS 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Max	Subgroup 2 T_A = 125°C Min	Max	Subgroup 3 T_A = -55°C Min	Max	Units
			1A	1B	2A	2B	2C	2Y	V_{ss}	1C	1Y	3Y	3A	3B	3C	V_{DD}								
t_{PHL}	3004 Fig. 5	121	13/	GND	GND	GND	GND		GND	GND	14/		GND	GND	GND	5.0 V	1Y	15	300	23	450	15	300	ns
		122	GND	13/						GND	14/						1Y							
		123		GND						13/	14/						1Y							
		124			13/			14/		GND							2Y							
		125			GND	13/		14/									2Y							
		126				GND	13/	14/									2Y							
		127					GND					14/	13/				3Y							
		128										14/	GND	13/			3Y							
		129										14/		GND	13/		3Y							
t_{PLH}		130	13/							GND	14/	GND			GND		1Y	21	410	31	615	21	410	
		131	GND	13/						GND	14/						1Y							
		132		GND						13/	14/						1Y							
		133			13/			14/		GND							2Y							
		134			GND	13/		14/									2Y							
		135				GND	13/	14/									2Y							
		136					GND					14/	13/				3Y							
		137					GND					14/	GND	13/			3Y							
		138					GND					14/	GND	GND	13/		3Y							

1/ Input pins not designated may be tied to V_{CC} or GND or may be left open provided they do not influence the outcome of the measurement. Output pins not designated may be tied to the loads or may be left open provided they do not influence the outcome of the measurement, except as follows: V_{IC} (pos) tests; the V_{ss} terminals shall be open. V_{IC} (neg) tests; the V_{DD} terminal shall be open. I_{SS} tests; the output terminals shall be open.

2/ When performing quiescent supply current measurements (I_{SS}), the meter shall be placed so that all currents flow thru the meter.

3/ V_{IL} = 1.1 V at 25°C, 0.85 V at 125°C, 1.35 V at -55°C. For device type 01, V_{IL} = 0.9 V at 25°C, 0.65 V at 125°C, and 0.95 V at -55°C.

4/ I_{OH} = -.10 mA at 25°C, -.10 mA at -125°C, -.10 mA at -55°C.

5/ I_{OH} = -.20 mA at 25°C, -.13 mA at 125°C, -.25 mA at -55°C.

6/ V_{IL} = 2.50 V at 25°C, 2.20 V at 125°C, 2.65 V at -55°C. For device type 01, V_{IL} = 2.25 V at 25°C, 1.95 V at 125°C, and 2.40 V at -55°C.

7/ V_{IH} = 3.60 V at 25°C, 3.65 V at 125°C, 3.95 V at -55°C. For device type 01, V_{IH} = 3.95 V at 25°C, 3.85 V at 125°C, and 4.05 V at -55°C.

8/ I_{OL} = .23 mA at 25°C, .23 mA at 125°C, .23 mA at -55°C.

9/ V_{IH} = 9.50 V at 25°C, 9.25 V at 125°C, 9.75 V at -55°C. For device type 01, V_{IH} = 10.25 V at 25°C, 10.00 V at 125°C, and 10.50 V at -55°C.

10/ I_{OL} = 0.8 mA at 25°C, 0.56 mA at 125°C, 1.0 mA at -55°C.

11/ *indicates the device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

12/ Capacitance bridge between measured terminal and V_{SS}; frequency = 1 MHz.

13/ Pulse conditions - See figure 5.

14/ Loading conditions - See figure 5.

15/ I_{OL} = .40 mA at 25°C, .28 mA at 125°C, .50 mA at -55°C.

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TABLE III. Group A inspection for device type 51.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	TERMINAL CONDITIONS 1/														Meas terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test No.	NC	NC	1A	1B	1C	1Y	V_{ss}	3A	3Y	2Y	2A	2B	2C	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1			1 mA											GND	1A		1.5					Vdc
		2				1 mA											1B							
		3					1 mA										1C							
		4								1 mA							3A							
		5											1 mA				2A							
		6												1 mA			2B							
		7													1 mA		2C							
V_{IC} (neg)		8			-1 mA				GND								1A		-6					
		9				-1 mA											1B							
		10					-1 mA										1C							
		11								-1 mA							3A							
		12											-1 mA				2A							
		13												-1 mA			2B							
		14													-1 mA		2C							
I_{DD} 2/	3005	15			18 V	GND	GND			18 V			18 V	GND	GND	18 V	V_{DD}		-25		-750			nA
		16			GND	18 V	GND			18 V			GND	18 V	GND		V_{DD}							
		17			GND	GND	18 V			18 V			GND	GND	18 V		V_{DD}							
		18			GND	GND	GND			GND			GND	GND	GND		V_{DD}							
V_{OH1}	3006	19			GND	GND	GND			15 V			15 V	15 V	15 V	15 V	1Y	14.95		14.95		14.95		Vdc
V_{OH2}	3006	20			15 V	15 V	15 V			15 V			GND	GND	GND		2Y							
V_{OH3}	3006	21			15 V	15 V	15 V			GND			15 V	15 V	15 V		3Y							
V_{OL1}	3007	22			15 V	GND	GND						GND	GND	GND		1Y		0.05		0.05		0.05	
		23			GND	15 V	GND						GND				1Y							
		24				GND	15 V						GND				1Y							
		25					GND						15 V				2Y							
		26											GND	15 V			2Y							
		27												GND	15 V		2Y							
		28								15 V				GND	GND		3Y							

See notes at end of table.

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TABLE III. Group A inspection for device type 51 – Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y Test No.	TERMINAL CONDITIONS 1/ 1 NC	2 NC	3 1A	4 1B	5 1C	6 1Y	7 V_{EE}	8 3A	9 3Y	10 2Y	11 2A	12 2B	13 2C	14 V_{CC}	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Subgroup 1 T_A = 25°C Max	Subgroup 2 T_A = 125°C Min	Subgroup 2 T_A = 125°C Max	Subgroup 3 T_A = -55°C Min	Subgroup 3 T_A = -55°C Max	Unit
V_{IH1}	3007	29			3.5 V	1.5 V	1.5 V		GND	GND			GND	GND	GND	5 V	1Y		0.5		0.5		0.5	Vdc
		30			1.5 V	3.5 V	1.5 V										1Y							
		31			1.5 V	1.5 V	3.5 V										1Y							
		32			GND	GND	GND						3.5 V	1.5 V	1.5 V		2Y							
		33											1.5 V	3.5 V	1.5 V		2Y							
		34											1.5 V	1.5 V	3.5 V		2Y							
		35								3.5 V			GND	GND	GND		3Y							
V_{IH2}		36			7.0 V	3.0 V	3.0 V			GND						10 V	1Y		1.0		1.0		1.0	
		37			3.0 V	7.0 V	3.0 V										1Y							
		38			3.0 V	3.0 V	7.0 V										1Y							
		39			GND	GND	GND						7.0 V	3.0 V	3.0 V		2Y							
		40											3.0 V	7.0 V	3.0 V		2Y							
		41											3.0 V	3.0 V	7.0 V		2Y							
		42								7.0 V			GND	GND	GND		3Y							
V_{IH3}		43			11 V	4.0 V	4.0 V			GND						15 V	1Y		1.5		1.5		1.5	
		44			4.0 V	11 V	4.0 V										1Y							
		45			4.0 V	4.0 V	11 V										1Y							
		46			GND	GND	GND						11 V	4.0 V	4.0 V		2Y							
		47											4.0 V	11 V	4.0 V		2Y							
		48											4.0 V	4.0 V	11 V		2Y							
		49								11 V			GND	GND	GND		3Y							
V_{IL1}		50			1.5 V	1.5 V	1.5 V			5.0 V			5.0 V	5.0 V	5.0 V	5 V	1Y	4.5		4.5		4.5		
V_{IL1}		51			5.0 V	5.0 V	5.0 V			5.0 V			1.5 V	1.5 V	1.5 V	5 V	2Y	4.5		4.5		4.5		
V_{IL1}		52			5.0 V	5.0 V	5.0 V			1.5 V			5.0 V	5.0 V	5.0 V	5 V	3Y	4.5		4.5		4.5		
V_{IL2}		53			3.0 V	3.0 V	3.0 V			10 V			10 V	10 V	10 V	10 V	1Y	9.0		9.0		9.0		
V_{IL2}		54			10 V	10 V	10 V			10 V			3.0 V	3.0 V	3.0 V	10 V	2Y	9.0		9.0		9.0		
V_{IL2}		55			10 V	10 V	10 V			3.0 V			10 V	10 V	10 V	10 V	3Y	9.0		9.0		9.0		
V_{IL3}		56			4.0 V	4.0 V	4.0 V			15 V			15 V	15 V	15 V	15 V	1Y	13.5		13.5		13.5		
V_{IL3}		57			15 V	15 V	15 V			15 V			4.0 V	4.0 V	4.0 V	15 V	2Y	13.5		13.5		13.5		
V_{IL3}		58			15 V	15 V	15 V			4.0 V			15 V	15 V	15 V	15 V	3Y	13.5		13.5		13.5		

See notes at end of table.

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TABLE III. Group A inspection for device type 51 – Continued.

Symbol	MIL-STD-883 method	Cases A, C, D X, Y Test No.	TERMINAL CONDITIONS 1/ 1 NC	2 NC	3 1A	4 1B	5 1C	6 1Y	7 V_{ss}	8 3A	9 3Y	10 2Y	11 2A	12 2B	13 2C	14 V_{DD}	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Max	Subgroup 2 T_A = 125°C Min	Max	Subgroup 3 T_A = -55°C Min	Max	Unit
I_{OL1}		59			5 V	GND	GND	0.4 V	GND	GND			GND	GND	GND	5 V	1Y	0.51		0.36		0.64		mA
		60			GND	5 V	GND	0.4 V					GND				1Y							
		61				GND	5 V	0.4 V					GND				1Y							
		62					GND					0.4 V	5 V				2Y							
		63										0.4 V	GND	5 V			2Y							
		64										0.4 V		GND	5 V		2Y							
		65								5 V	0.4 V				GND		3Y							
I_{OL2}		66			15 V			1.5 V		GND						15 V	1Y	3.4		2.4		4.2		
		67			GND	15 V		1.5 V									1Y							
		68				GND	15 V	1.5 V									1Y							
		69					GND					1.5 V	15 V				2Y							
		70										1.5 V	GND	15 V			2Y							
		71										1.5 V	GND	GND	15 V		2Y							
		72								15 V	1.5 V		GND	GND	GND		3Y							
I_{OH1}		73						4.6 V		5 V			5 V	5 V	5 V	5 V	1Y	-0.51		-0.36		-0.64		
I_{OH2}		74			5 V	5 V	5 V			5 V		4.6 V	GND	GND	GND	5 V	2Y	-0.51		-0.36		-0.64		
I_{OH3}		75			5 V	5 V	5 V			GND	4.6 V		5 V	5 V	5 V	5 V	3Y	-0.51		-0.36		-0.64		
I_{OH4}		76			GND	GND	GND	13.5 V		15 V			15 V	15 V	15 V	15 V	1Y	-3.4		-2.4		-4.2		
I_{OH5}		77			15 V	15 V	15 V			15 V		13.5 V	GND	GND	GND	15 V	2Y	-3.4		-2.4		-4.2		
I_{OH6}		78			15 V	15 V	15 V			GND	13.5 V		15 V	15 V	15 V	15 V	3Y	-3.4		-2.4		-4.2		
I_{IL} 3/	3009	79			GND	GND	GND						GND	GND	GND	18 V	All inputs together		-7.0					nA
		80															1A		-1.0		-.45			
		81															1B							
		82															1C							
		83															2A							
		84															2B							
		85															2C							
		86															3A							

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TABLE III. Group A inspection for device type 51 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	TERMINAL CONDITIONS 1/														Test limits						Units
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Subgroup 1 T_A = 25°C Min	Subgroup 1 T_A = 25°C Max	Subgroup 2 T_A = 125°C Min	Subgroup 2 T_A = 125°C Max	Subgroup 3 T_A = -55°C Min	Subgroup 3 T_A = -55°C Max	
I_{IH}	3010	87	NC	NC	1A 18.0 V	1B 18.0 V	1C 18.0 V	1Y	V_{CC} GND	3A 18.0 V	2Y	2A 18.0 V	2B 18.0 V	2C 18.0 V	V_{CC} 18 V	All inputs together		7.0		1.0			nA
		88			18.0 V	GND	GND			GND		GND	GND	GND		1A				45			
		89			GND	18.0 V	GND									1B							
		90				GND	18.0 V									1C							
		91				↓	GND						18.0 V			2A							
		92											GND	18.0 V		2B							
		93											GND	GND	18.0 V	2C							
		94								18.0 V					GND	3A							
																	Subgroup 4 T_A = 25°C						
C_I	3012	95	NC	NC	1A ↓	1B ↓	1C ↓	1Y	V_{CC} GND	3A ↓	2Y	2A ↓	2B ↓	2C ↓	V_{CC} GND	V_{CC} GND			12					pF
		96															1A						
		97															1B						
		98															1C						
		99															2A						
		100															2B						
		101								↓							2C						
																	3A						
																	Subgroup 9 T_A = 25°C			Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C	
I_{OHL}	3003	102	NC	NC	↓	↓	GND	↓	GND	GND	↓	GND	GND	GND	GND	5.0 V	1A to 1Y	11	240	16	315	11	230	ns
	Fig. 5	103			GND	↓	GND						GND	GND			1B to 1Y						
		104				GND	↓						GND	↓			1C to 1Y						
		105					GND				↓		↓	↓			2A to 2Y						
		106									↓		GND	↓			2B to 2Y						
		107									↓		↓	GND	↓		2C to 2Y						
		108								↓	↓			↓	GND		3A to 3Y						

See notes at end of table.

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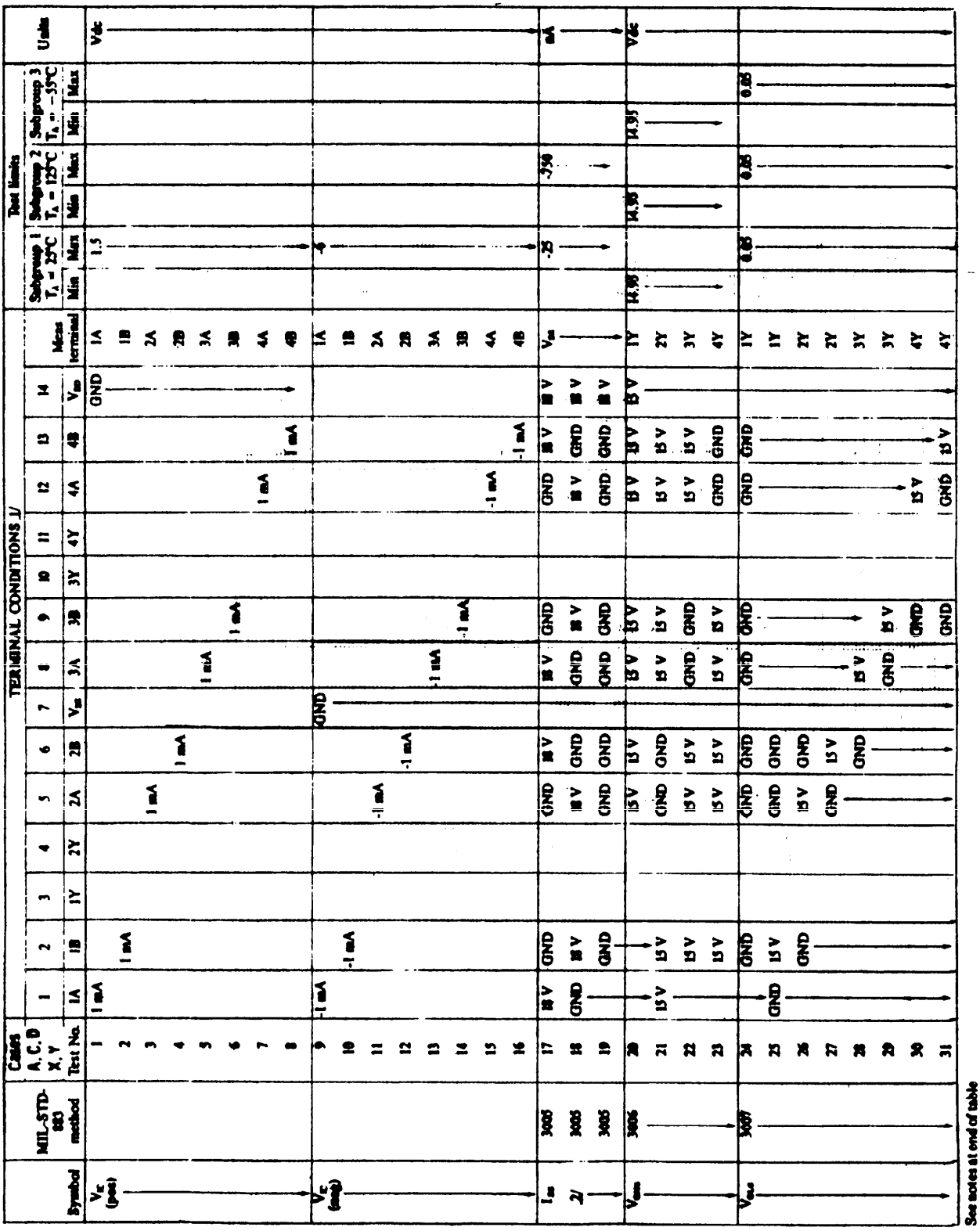
TABLE III. Group A inspection for device type 51 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D X, Y Test No.	TERMINAL CONDITIONS 1/ 1 NC	2 NC	3 1A	4 1B	5 1C	6 1Y	7 V_{EE}	8 3A	9 3Y	10 2Y	11 2A	12 2B	13 2C	14 V_{DD}	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Subgroup 1 T_A = 25°C Max	Subgroup 2 T_A = 125°C Min	Subgroup 2 T_A = 125°C Max	Subgroup 3 T_A = -55°C Min	Subgroup 3 T_A = -55°C Max	Units
t_{PLH}	3003 Fig. 5	109			5/	GND	GND	6/	GND	GND			GND	GND	GND	5.0 V	1A to 1Y	14	275	21	415	14	275	ns
		110			GND	5/	GND	6/									1B to 1Y							
		111				GND	5/	6/									1C to 1Y							
		112					GND					6/	5/				2A to 2Y							
		113										6/	GND	5/			2B to 2Y							
		114										6/		GND	5/		2C to 2Y							
		115								5/	6/				GND		3A to 3Y							
t_{PHL}	3004 Fig. 5	116			5/			6/		GND							1Y	15	300	23	450	15	300	
		117			GND	5/		6/									1Y							
		118				GND	5/	6/									1Y							
		119					GND					6/	5/				2Y							
		120										6/	GND	5/			2Y							
		121										6/		GND	5/		2Y							
		122								5/	6/				GND		3Y							
t_{PLH}		123			5/			6/		GND							1Y	23	450	34	675	23	450	
		124			GND	5/		6/									1Y							
		125				GND	5/	6/									1Y							
		126					GND					6/	5/				2Y							
		127										6/	GND	5/			2Y							
		128										6/		GND	5/		2Y							
		129								5/	6/			GND	GND		3Y							

See notes at end of table.

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TABLE III. Group A inspection for device type 52.



Symbol	MIL-STD-883 method	Cases A, C, D X, Y Test No.	TERMINAL CONDITIONS 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Subgroup 1 T_A = 25°C Max	Subgroup 2 T_A = 125°C Min	Subgroup 2 T_A = 125°C Max	Subgroup 3 T_A = -55°C Min	Subgroup 3 T_A = -55°C Max	Units
V_{IC} (pos)		1	1A: 1 mA	1B: 1 mA	1Y	2Y	2A: 1 mA	2B: 1 mA	V_{cc}: GND	3A: 1 mA	3B: 1 mA	3Y	4Y	4A: 1 mA	4B: 1 mA	V_{cc}	1A		1.5					Vdc
		2	1A: 1 mA	1B: 1 mA	1Y	2Y	2A: 1 mA	2B: 1 mA	V_{cc}: GND	3A: 1 mA	3B: 1 mA	3Y	4Y	4A: 1 mA	4B: 1 mA	V_{cc}	1B		1.5					Vdc
		3	1A: 1 mA	1B: 1 mA	1Y	2Y	2A: 1 mA	2B: 1 mA	V_{cc}: GND	3A: 1 mA	3B: 1 mA	3Y	4Y	4A: 1 mA	4B: 1 mA	V_{cc}	2A		1.5					Vdc
		4	1A: 1 mA	1B: 1 mA	1Y	2Y	2A: 1 mA	2B: 1 mA	V_{cc}: GND	3A: 1 mA	3B: 1 mA	3Y	4Y	4A: 1 mA	4B: 1 mA	V_{cc}	2B		1.5					Vdc
		5	1A: 1 mA	1B: 1 mA	1Y	2Y	2A: 1 mA	2B: 1 mA	V_{cc}: GND	3A: 1 mA	3B: 1 mA	3Y	4Y	4A: 1 mA	4B: 1 mA	V_{cc}	3A		1.5					Vdc
		6	1A: 1 mA	1B: 1 mA	1Y	2Y	2A: 1 mA	2B: 1 mA	V_{cc}: GND	3A: 1 mA	3B: 1 mA	3Y	4Y	4A: 1 mA	4B: 1 mA	V_{cc}	3B		1.5					Vdc
		7	1A: 1 mA	1B: 1 mA	1Y	2Y	2A: 1 mA	2B: 1 mA	V_{cc}: GND	3A: 1 mA	3B: 1 mA	3Y	4Y	4A: 1 mA	4B: 1 mA	V_{cc}	4A		1.5					Vdc
		8	1A: 1 mA	1B: 1 mA	1Y	2Y	2A: 1 mA	2B: 1 mA	V_{cc}: GND	3A: 1 mA	3B: 1 mA	3Y	4Y	4A: 1 mA	4B: 1 mA	V_{cc}	4B		1.5					Vdc
V_{IC} (neg)		9	1A: -1 mA	1B: -1 mA	1Y	2Y	2A: -1 mA	2B: -1 mA	V_{cc}: GND	3A: -1 mA	3B: -1 mA	3Y	4Y	4A: -1 mA	4B: -1 mA	V_{cc}	1A		-4					Vdc
		10	1A: -1 mA	1B: -1 mA	1Y	2Y	2A: -1 mA	2B: -1 mA	V_{cc}: GND	3A: -1 mA	3B: -1 mA	3Y	4Y	4A: -1 mA	4B: -1 mA	V_{cc}	1B		-4					Vdc
		11	1A: -1 mA	1B: -1 mA	1Y	2Y	2A: -1 mA	2B: -1 mA	V_{cc}: GND	3A: -1 mA	3B: -1 mA	3Y	4Y	4A: -1 mA	4B: -1 mA	V_{cc}	2A		-4					Vdc
		12	1A: -1 mA	1B: -1 mA	1Y	2Y	2A: -1 mA	2B: -1 mA	V_{cc}: GND	3A: -1 mA	3B: -1 mA	3Y	4Y	4A: -1 mA	4B: -1 mA	V_{cc}	2B		-4					Vdc
		13	1A: -1 mA	1B: -1 mA	1Y	2Y	2A: -1 mA	2B: -1 mA	V_{cc}: GND	3A: -1 mA	3B: -1 mA	3Y	4Y	4A: -1 mA	4B: -1 mA	V_{cc}	3A		-4					Vdc
		14	1A: -1 mA	1B: -1 mA	1Y	2Y	2A: -1 mA	2B: -1 mA	V_{cc}: GND	3A: -1 mA	3B: -1 mA	3Y	4Y	4A: -1 mA	4B: -1 mA	V_{cc}	3B		-4					Vdc
		15	1A: -1 mA	1B: -1 mA	1Y	2Y	2A: -1 mA	2B: -1 mA	V_{cc}: GND	3A: -1 mA	3B: -1 mA	3Y	4Y	4A: -1 mA	4B: -1 mA	V_{cc}	4A		-4					Vdc
		16	1A: -1 mA	1B: -1 mA	1Y	2Y	2A: -1 mA	2B: -1 mA	V_{cc}: GND	3A: -1 mA	3B: -1 mA	3Y	4Y	4A: -1 mA	4B: -1 mA	V_{cc}	4B		-4					Vdc
I_{IH} 2/	3005	17	1A: 18 V	1B: GND	1Y	2Y	2A: GND	2B: 18 V	V_{cc}: GND	3A: 18 V	3B: GND	3Y	4Y	4A: GND	4B: 18 V	V_{cc}	1Y		-.25		-.350			mA
	3005	18	1A: GND	1B: 18 V	1Y	2Y	2A: 18 V	2B: GND	V_{cc}: GND	3A: GND	3B: 18 V	3Y	4Y	4A: 18 V	4B: GND	V_{cc}	2Y		-.25		-.350			mA
	3005	19	1A: GND	1B: GND	1Y	2Y	2A: GND	2B: GND	V_{cc}: GND	3A: GND	3B: GND	3Y	4Y	4A: GND	4B: 18 V	V_{cc}	3Y		-.25		-.350			mA
V_{OHH}	3006	20	1A: 15 V	1B: 15 V	1Y	2Y	2A: 15 V	2B: 15 V	V_{cc}: GND	3A: 15 V	3B: 15 V	3Y	4Y	4A: 15 V	4B: 15 V	V_{cc}	4Y	14.95		14.95		14.95		Vdc
		21	1A: 15 V	1B: GND	1Y	2Y	2A: GND	2B: GND	V_{cc}: GND	3A: 15 V	3B: 15 V	3Y	4Y	4A: 15 V	4B: 15 V	V_{cc}	1Y	14.95		14.95		14.95		Vdc
		22	1A: 15 V	1B: 15 V	1Y	2Y	2A: 15 V	2B: 15 V	V_{cc}: GND	3A: GND	3B: GND	3Y	4Y	4A: 15 V	4B: 15 V	V_{cc}	2Y	14.95		14.95		14.95		Vdc
		23	1A: 15 V	1B: 15 V	1Y	2Y	2A: 15 V	2B: 15 V	V_{cc}: GND	3A: 15 V	3B: GND	3Y	4Y	4A: GND	4B: 15 V	V_{cc}	3Y	14.95		14.95		14.95		Vdc
V_{OLL}	3007	24	1A: GND	1B: GND	1Y	2Y	2A: GND	2B: GND	V_{cc}: GND	3A: GND	3B: GND	3Y	4Y	4A: GND	4B: GND	V_{cc}	1Y		0.05		0.05		0.05	Vdc
		25	1A: GND	1B: GND	1Y	2Y	2A: GND	2B: GND	V_{cc}: GND	3A: GND	3B: GND	3Y	4Y	4A: GND	4B: GND	V_{cc}	1Y		0.05		0.05		0.05	Vdc
		26	1A: GND	1B: GND	1Y	2Y	2A: GND	2B: GND	V_{cc}: GND	3A: GND	3B: GND	3Y	4Y	4A: GND	4B: GND	V_{cc}	2Y		0.05		0.05		0.05	Vdc
		27	1A: GND	1B: GND	1Y	2Y	2A: GND	2B: GND	V_{cc}: GND	3A: GND	3B: GND	3Y	4Y	4A: GND	4B: GND	V_{cc}	2Y		0.05		0.05		0.05	Vdc
		28	1A: GND	1B: GND	1Y	2Y	2A: GND	2B: GND	V_{cc}: GND	3A: GND	3B: GND	3Y	4Y	4A: GND	4B: GND	V_{cc}	3Y		0.05		0.05		0.05	Vdc
		29	1A: GND	1B: GND	1Y	2Y	2A: GND	2B: GND	V_{cc}: GND	3A: GND	3B: GND	3Y	4Y	4A: GND	4B: GND	V_{cc}	3Y		0.05		0.05		0.05	Vdc
		30	1A: GND	1B: GND	1Y	2Y	2A: GND	2B: GND	V_{cc}: GND	3A: GND	3B: GND	3Y	4Y	4A: GND	4B: GND	V_{cc}	4Y		0.05		0.05		0.05	Vdc
		31	1A: GND	1B: GND	1Y	2Y	2A: GND	2B: GND	V_{cc}: GND	3A: GND	3B: GND	3Y	4Y	4A: GND	4B: GND	V_{cc}	4Y		0.05		0.05		0.05	Vdc

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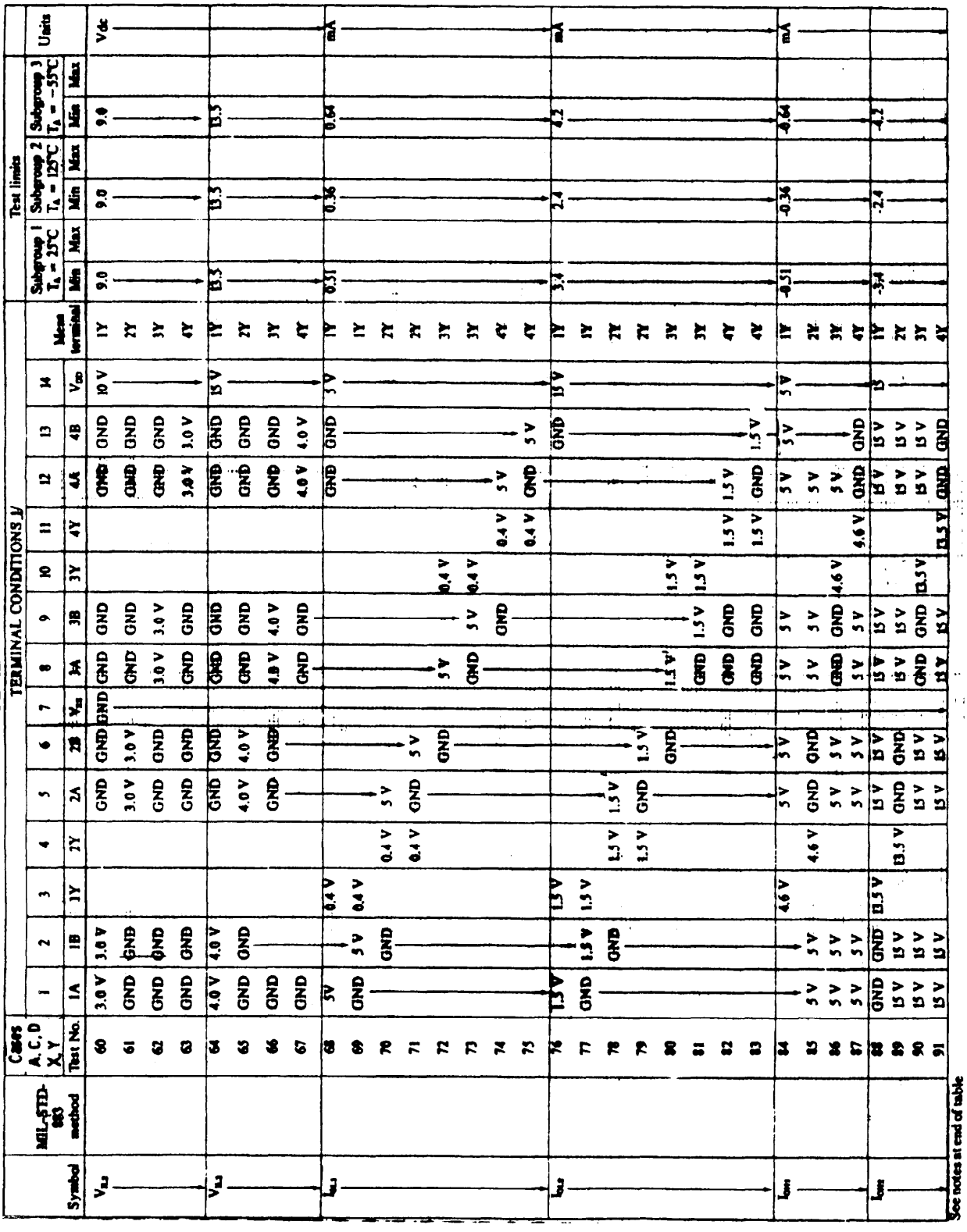
TABLE III. Group A inspection for device type 52 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y Test No.	TERMINAL CONDITIONS 1/ 1 (1A)	2 (1B)	3 (1Y)	4 (2Y)	5 (2A)	6 (2B)	7 (V_{EE})	8 (3A)	9 (3B)	10 (3Y)	11 (4Y)	12 (4A)	13 (4B)	14 (V_{CC})	Meas. terminal	Test Limits: Subgroup 1 T_A = 25°C Min	Subgroup 1 T_A = 25°C Max	Subgroup 2 T_A = 125°C Min	Subgroup 2 T_A = 125°C Max	Subgroup 3 T_A = -55°C Min	Subgroup 3 T_A = -55°C Max	Units
V_{IH1}		32	3.5 V	1.5 V			GND	GND	GND	GND	GND			GND	GND	5 V	1Y		0.5		0.5		0.5	Vdc
		33	1.5 V	3.5 V			3.5 V	1.5 V									2Y							
		34	GND	GND			1.5 V	3.5 V									3Y							
		35					GND	GND		3.5 V	1.5 V						4Y							
		36								1.5 V	3.5 V			3.5 V	1.5 V									
		37								GND	GND			1.5 V	3.5 V									
		38																						
		39																						
V_{IH2}		40	7.0 V	3.0 V			GND	GND	GND	GND	GND			GND	GND	10 V	1Y		1.0		1.0		1.0	
		41	3.0 V	7.0 V			GND	GND									2Y							
		42	GND	GND			7.0 V	3.0 V									3Y							
		43					3.0 V	7.0 V		7.0 V	3.0 V						4Y							
		44					GND	GND		3.0 V	7.0 V													
		45								GND	GND			7.0 V	3.0 V									
		46												3.0 V	7.0 V									
		47																						
V_{IH3}		48	11 V	4.5 V			GND	GND	GND	GND	GND			GND	GND	15 V	1Y		1.5		1.5		1.5	
		49	4.0 V	11 V			11 V	4.0 V									2Y							
		50	GND	GND			4.0 V	11 V									3Y							
		51					11 V	4.0 V		11 V	4.0 V						4Y							
		52					GND	GND		4.0 V	11 V													
		53								GND	GND			11 V	4.0 V									
		54												4.0 V	11 V									
		55																						
V_{IL}		56	1.5 V	1.5 V			GND	1.5 V	GND	GND	GND			GND	GND	5 V	1Y	4.5		4.5		4.5		
		57	GND	GND			1.5 V	GND									2Y							
		58	GND	GND			GND	GND		1.5 V	1.5 V						3Y							
		59	GND	GND			GND	GND		GND	GND			1.5 V	1.5 V		4Y							

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TABLE III. Group A inspection for device type 52 - Continued.



Symbol	MIL-STD-883 method	Cases A, C, D X, Y Test No.	TERMINAL CONDITIONS 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Subgroup 1 T_A = 25°C Max	Subgroup 2 T_A = 125°C Min	Subgroup 2 T_A = 125°C Max	Subgroup 3 T_A = -55°C Min	Subgroup 3 T_A = -55°C Max	Units
V_{IL3}		60	3.0 V	3.0 V			GND	GND	GND	GND	GND			GND	GND	10 V	1Y	9.0		9.0		9.0		Vdc
		61	GND	GND			3.0 V	3.0 V		GND	GND			GND	GND		2Y							
		62	GND	GND			GND	GND		3.0 V	3.0 V			GND	GND		3Y							
		63	GND	GND			GND	GND		GND	GND			3.0 V	3.0 V		4Y							
V_{IL3}		64	4.0 V	4.0 V			GND	GND		GND	GND			GND	GND	15 V	1Y	13.5		13.5		13.5		
		65	GND	GND			4.0 V	4.0 V		GND	GND			GND	GND		2Y							
		66	GND	GND			GND	GND		4.0 V	4.0 V			GND	GND		3Y							
		67	GND	GND			GND	GND		GND	GND			4.0 V	4.0 V		4Y							
I_{IL3}		68	5V		0.4 V		GND	GND		GND	GND			GND	GND	5 V	1Y	0.51		0.36		0.64		mA
		69	GND	5 V	0.4 V		5 V			GND	GND						1Y							
		70		GND		0.4 V		5 V				0.4 V	0.4 V				2Y							
		71				0.4 V	GND										2Y							
		72						GND		5 V	0.4 V						3Y							
		73								GND	5 V	0.4 V					3Y							
		74									GND		0.4 V	5 V	GND		4Y							
		75											0.4 V	GND	5 V		4Y							
I_{IL3}		76	1.5 V		1.5 V		1.5 V	1.5 V		1.5 V				GND	GND	15 V	1Y	3.4		2.4		4.2		
		77	GND	1.5 V	1.5 V												1Y							
		78		GND		1.5 V	1.5 V										2Y							
		79				1.5 V	GND	1.5 V									2Y							
		80						GND		1.5 V	1.5 V						3Y							
		81								GND	1.5 V	1.5 V					3Y							
		82								GND	GND		1.5 V	1.5 V			4Y							
		83								GND	GND		1.5 V	GND	1.5 V		4Y							
I_{IH3}		84			4.6 V		5 V	5 V		5 V	5 V			5 V	5 V	5 V	1Y	-0.51		-0.36		-0.64		mA
		85	5 V	5 V		4.6 V	GND	GND		5 V	5 V			5 V			2Y							
		86	5 V	5 V			5 V	5 V		GND	GND	4.6 V		5 V			3Y							
		87	5 V	5 V			5 V	5 V		5 V	5 V		4.6 V	GND	GND		4Y							
I_{IH3}		88	GND	GND	13.5 V		15 V	15 V		15 V	15 V			15 V	15 V	15	1Y	-3.4		-2.4		-4.2		
		89	15 V	15 V		13.5 V	GND	GND		15 V	15 V			15 V	15 V		2Y							
		90	15 V	15 V			15 V	15 V		GND	GND	13.5 V		15 V	15 V		3Y							
		91	15 V	15 V			15 V	15 V		15 V	15 V		13.5 V	GND	GND		4Y							

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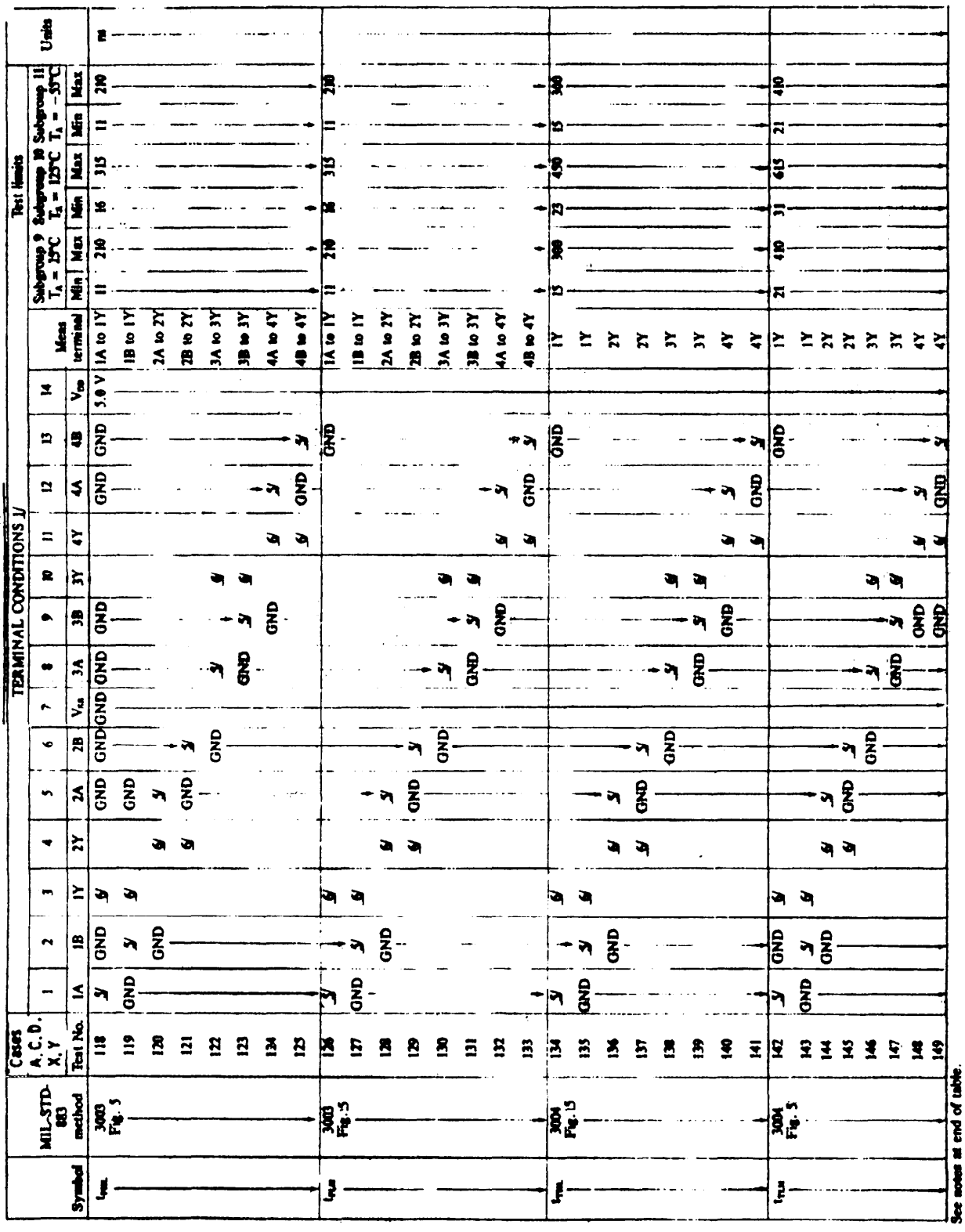
TABLE III. Group A inspection for device type 52 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D X, Y	TERMINAL CONDITIONS 1/														Meas terminal	Test limits						Units
																		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
			1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}								
I_{IL} 3/	3009	92	GND	GND			GND	GND	GND	GND	GND			GND	GND	18.0 V	All inputs together		-8.0					nA
		93															1A		-1.0		-45			
		94															1B							
		95															2A							
		96															2B							
		97															3A							
		98															3B							
		99															4A							
		100															4B							
I_{IH} 3/	3010	101	18.0 V	18.0 V			18.0 V	18.0 V		18.0 V	18.0 V			18.0 V	18.0 V		All inputs together		8.0					
		102	18.0 V	GND			GND	GND		GND	GND			GND	GND		1A		+1.0		45			
		103	GND	18.0 V			GND	GND									1B							
		104		GND			18.0 V	GND									2A							
		105					GND	18.0 V									2B							
		106						GND		18.0 V							3A							
		107								GND	18.0 V						3B							
		108								GND	GND			18.0 V			4A							
		109								GND	GND			GND	18.0 V		4B							
																		Subgroup 4 T_A = 25°C						
C_I	3012	110	4/						GND							GND	1A		12					pF
		111		4/													1B							
		112					4/										2A							
		113						4/									2B							
		114								4/							3A							
		115									4/						3B							
		116												4/			4A							
		117													4/		4B							

See notes at end of table.

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TABLE III. Group A inspection for device type 52 – Continued.



Symbol	MIL-STD-883 method	Cases A, C, D, X, Y Test No.	TERMINAL CONDITIONS 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Test Limits Subgroup 9 T_A = 25°C Min	Subgroup 9 T_A = 25°C Max	Subgroup 10 Subgroup 11 T_A = 125°C Min	Subgroup 10 Subgroup 11 T_A = 125°C Max	T_A = −55°C Min	T_A = −55°C Max	Units
t_{PHL}	3003 Fig. 5	118	1A	1B	1Y	2Y	2A	2B	V_{CC}	3A	3B	3Y	4Y	4A	4B	V_{CC}	1A to 1Y	11	210	16	315	11	210	ns
		119	GND	GND	↓	↓	GND	↓	GND	↓	↓	↓	↓	↓	↓	5.0 V	1B to 1Y							
		120	↓	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	2A to 2Y							
		121	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	2B to 2Y							
		122	↓	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	3A to 3Y							
		123	↓	↓	↓	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	↓	3B to 3Y							
		134	↓	↓	↓	↓	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	4A to 4Y							
		125	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	GND	↓	↓	4B to 4Y							
t_{PLH}	3003 Fig. 5	126	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	GND	↓	1A to 1Y	11	210	16	315	11	210	
		127	GND	↓	↓	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	1B to 1Y							
		128	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	2A to 2Y							
		129	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	2B to 2Y							
		130	↓	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	3A to 3Y							
		131	↓	↓	↓	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	↓	3B to 3Y							
		132	↓	↓	↓	↓	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	4A to 4Y							
		133	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	GND	↓	↓	4B to 4Y							
t_{PHL}	3004 Fig. 15	134	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	GND	↓	1Y	15	300	23	450	15	300	
		135	GND	↓	↓	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	1Y							
		136	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	2Y							
		137	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	2Y							
		138	↓	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	3Y							
		139	↓	↓	↓	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	↓	3Y							
		140	↓	↓	↓	↓	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	4Y							
		141	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	GND	↓	↓	4Y							
t_{PLH}	3004 Fig. 5	142	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	GND	↓	1Y	21	410	31	615	21	410	
		143	GND	↓	↓	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	1Y							
		144	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	2Y							
		145	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	2Y							
		146	↓	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	3Y							
		147	↓	↓	↓	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	↓	3Y							
		148	↓	↓	↓	↓	↓	↓	↓	↓	GND	↓	↓	↓	↓	↓	4Y							
		149	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	GND	↓	↓	4Y							

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TABLE ... A inspection for device type 53.

Symbol	MIL-STD-883 method	Cases K, C, D X, Y Test No.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Max	Subgroup 2 T_A = 125°C Min	Max	Subgroup 3 T_A = −55°C Min	Max	Units
V_{cc} (pos)		1	1Y	1 mA	1B	1C	1D	NC	V_{ee}	NC	2A	2B	2C	2D	2Y	V_{cc}	1A		1.5					Vdc
		2															1B							
		3															1C							
		4															1D							
		5															2A							
		6															2B							
		7															2C							
		8															2D							
V_{cc} (neg)		9	1A						GND								1A		-6					
		10															1B							
		11															1C							
		12															1D							
		13															2A							
		14															2B							
		15															2C							
		16															2D							
I_{ee}	3005	17	GND	GND	GND	GND	GND		GND		GND	GND	GND	GND		15 V	V_{ee}		-25		-750			mA
		18	GND	GND	GND	GND	18 V		GND		18 V	GND												
		19	GND	GND	GND	18 V	GND		GND		GND	18 V												
		20	GND	GND	18 V	GND	GND		GND		GND	GND	18 V											
		21	GND	18 V	GND	GND	GND		GND		GND	GND	GND	18 V										
V_{oss}	3006	22	GND	GND	GND	GND	GND		GND		15 V	15 V	15 V	15 V		15 V	1Y	14.95		14.95		14.95		Vdc
		23	15 V	15 V	15 V	15 V	15 V		GND		GND	GND	GND	GND		GND	2Y	14.95		14.95				
V_{oss}	3006	24	15 V	GND	GND	GND	GND		GND		GND	GND	GND	GND		GND	1Y		0.05		0.05		0.05	
V_{out}	3007	25	GND	GND	15 V	GND	GND		GND		GND	GND	GND	GND		GND	1Y							
		26	GND	15 V	GND	15 V	GND		GND		GND	GND	GND	GND		GND	1Y							
		27	GND	GND	GND	GND	15 V		GND		15 V	GND	15 V	GND		GND	1Y							
		28	GND	GND	GND	15 V	GND		GND		GND	15 V	GND	GND		GND	2Y							
		29	GND	GND	15 V	GND	GND		GND		GND	GND	15 V	GND		GND	2Y							
		30	GND	15 V	GND	GND	GND		GND		GND	GND	GND	15 V		GND	2Y							
		31	15 V	GND	GND	GND	GND		GND		GND	GND	GND	GND		15 V	2Y							

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TABLE III. Group A inspection for device type 53 – Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	TERMINAL CONDITIONS 1/															Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test No.	1Y	1A	1B	1C	1D	NC	V_{EE}	NC	2A	2B	2C	2D	2Y	Vdc		Min	Max	Min	Max	Min	Max	
V_{IH1}		32		3.5 V	1.5 V	1.5 V	1.5 V		GND		1.5 V	1.5 V	1.5 V	1.5 V		5 V	1Y		0.5		0.5		0.5	Vdc
		33		1.5 V	3.5 V	1.5 V	1.5 V										1Y							
		34			1.5 V	3.5 V	1.5 V										1Y							
		35				1.5 V	3.5 V										1Y							
		36					1.5 V				3.5 V						2Y							
		37									1.5 V	3.5 V					2Y							
		38									1.5 V	1.5 V	3.5 V				2Y							
		39									1.5 V	1.5 V	1.5 V	3.5 V			2Y							
V_{IH2}		40		7.0 V	3.0 V	3.0 V	3.0 V				3.0 V	3.0 V	3.0 V	3.0 V		10 V	1Y		1.0		1.0		1.0	
		41		3.0 V	7.0 V	3.0 V	3.0 V										1Y							
		42			3.0 V	7.0 V	3.0 V										1Y							
		43				3.0 V	7.0 V										1Y							
		44					3.0 V				7.0 V						2Y							
		45									3.0 V	7.0 V					2Y							
		46									3.0 V	3.0 V	7.0 V				2Y							
		47									3.0 V	3.0 V	3.0 V	7.0 V			2Y							
V_{IH3}		48		11 V	4.0 V	4.0 V	4.0 V				4.0 V	4.0 V	4.0 V	4.0 V		15 V	1Y		1.5		1.5		1.5	
		49		4.0 V	11 V	4.0 V											1Y							
		50			4.0 V	11 V											1Y							
		51				4.0 V	11 V										1Y							
		52					4.0 V				11 V						2Y							
		53									4.0 V	11 V					2Y							
		54									4.0 V	4.0 V	11 V				2Y							
		55									4.0 V	4.0 V	4.0 V	11 V			2Y							
V_{IL1}		56		1.5 V	1.5 V	1.5 V	1.5 V				5 V	5 V	5 V	5 V		5 V	1Y	4.5		4.5		4.5		
V_{IL1}		57		5 V	5 V	5 V	5 V				1.5 V	1.5 V	1.5 V	1.5 V		5 V	2Y	4.5		4.5		4.5		
V_{IL2}		58		3.0 V	3.0 V	3.0 V	3.0 V				10 V	10 V	10 V	10 V		10 V	1Y	9.0		9.0		9.0		
V_{IL2}		59		10 V	10 V	10 V	10 V				3.0 V	3.0 V	3.0 V	3.0 V		10 V	2Y	9.0		9.0		9.0		
V_{IL3}		60		4.0 V	4.0 V	4.0 V	4.0 V				15 V	15 V	15 V	15 V		15 V	1Y	13.5		13.5		13.5		
V_{IL3}		61		15 V	15 V	15 V	15 V				4.0 V	4.0 V	4.0 V	4.0 V		15 V	2Y	13.5		13.5		13.5		

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TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D X, Y Test No.	TERMINAL CONDITIONS 1/ 1 1Y	2 1A	3 1B	4 1C	5 1D	6 NC	7 V_{EE}	8 NC	9 2A	10 2B	11 2C	12 2D	13 2Y	14 V_{CC}	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Subgroup 1 T_A = 25°C Max	Subgroup 2 T_A = 125°C Min	Subgroup 2 T_A = 125°C Max	Subgroup 3 T_A = -55°C Min	Subgroup 3 T_A = -55°C Max	Units
I_{IL1}		62	0.4 V	5 V	GND	GND	GND		GND		GND	GND	GND	GND		5 V	1Y	0.51		0.36		0.64		mA
		63		GND	5 V	GND	GND				GND	GND	GND				1Y							
		64			GND	5 V	GND				GND	GND	GND				1Y							
		65				GND	5 V				5 V	GND	GND				1Y							
		66					GND				5 V	GND	GND		0.4 V		2Y							
		67									GND	5 V	5 V				2Y							
		68										GND	5 V	5 V			2Y							
		69											GND				2Y							
I_{IL2}		70	1.5 V	15 V					GND		GND	GND	GND	GND		15 V	1Y	3.4		2.4		4.2		
		71		GND													1Y							
		72					15 V										1Y							
		73									15 V				1.5 V		2Y							
		74										15 V					2Y							
		75									GND	15 V	15 V				2Y							
		76									GND	GND	15 V				2Y							
		77									GND	GND	GND	15 V			2Y							
I_{IH1}		78	4.6 V	5 V							5 V	5 V	5 V	5 V	5 V	5 V	1Y	-0.51		-0.36		-0.64		
I_{IH1}		79	5 V	5 V							GND	GND	GND	GND	4.6 V	5 V	2Y	-0.51		-0.36		-0.64		
I_{IH2}		80	13.5 V	GND	GND	GND	GND				15 V	15 V	15 V	15 V	15 V	15 V	1Y	-3.4		-2.4		-4.2		
I_{IH2}		81	15 V	15 V	15 V	15 V	15 V				GND	GND	GND	GND	13.5 V	15 V	2Y	-3.4		-2.4		-4.2		
I_X	3009	82		GND	GND	GND	GND		GND		GND	GND	GND	GND		18.0 V	All inputs together		-3.0					mA
I_Y		83															1A		-1.0		-45			
		84															1B							
		85															1C							
		86															1D							
		87															2A							
		88															2B							
		89															2C							
		90															2D							

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TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y Test No.	TERMINAL CONDITIONS 1/ 1 1Y	2 1A	3 1B	4 1C	5 1D	6 NC	7 V_{EE}	8 NC	9 2A	10 2B	11 2C	12 2D	13 2Y	14 V_{CC}	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Max	Subgroup 2 T_A = 125°C Min	Max	Subgroup 3 T_A = -55°C Min	Max	Units
I_{IH} 2/	3010	91		18.0 V	18.0 V	18.0 V	18.0 V		GND		18.0 V	18.0 V	18.0 V	18.0 V		18.0 V	All inputs together		8.0					nA
		92		18.0 V	GND	GND	GND				GND	GND	GND	GND			1A		1.0		45			
		93		GND	18.0 V	GND	GND										1B							
		94			GND	18.0 V	GND										1C							
		95				GND	18.0 V										1D							
		96					GND				18.0 V						2A							
		97									GND	18.0 V					2B							
		98									GND	GND	18.0 V				2C							
		99									GND	GND	GND	18.0 V			2D							
																		Subgroup 4 T_A = 25°C						
C_i	3012	100		4/					GND							GND	1A		12					pF
		101			4/												1B							
		102				4/											1C							
		103					4/										1D							
		104									4/						2A							
		105										4/					2B							
		106											4/				2C							
		107												4/			2D							
																		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
t_{PHL}, t_{PLH}	3003 Fig. 5	108	5/	5/	GND	GND	GND		GND		GND	GND	GND	GND		5.0 V	1A to 1Y	11	210	16	315	11	210	ns
		109		GND	5/	GND	GND										1B to 1Y							
		110			GND	5/	GND										1C to 1Y							
		111				GND	5/										1D to 1Y							
		112					GND				5/				5/		2A to 2Y							
		113									GND	5/					2B to 2Y							
		114									GND	GND	5/				2C to 2Y							
		115									GND	GND	GND	5/			2D to 2Y							

See notes at end of table.

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TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	TERMINAL CONDITIONS 1/														Meas terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test No.	1Y	1A	1B	1C	1D	NC	V_{ss}	NC	2A	2B	2C	2D	2Y	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{PLH}	3003 Fig. 5	116	↓	↓	GND	GND	GND		GND		GND	GND	GND	GND		5.0 V	1A to 1Y	11	210	16	315	12	210	ns
		117		GND	↓	GND											1B to 1Y							
		118			GND	↓											1C to 1Y							
		119				GND	↓										1D to 1Y							
		120					GND				↓				↓		2A to 2Y							
		121									GND	↓					2B to 2Y							
		122										GND	↓				2C to 2Y							
		123											GND	↓			2D to 2Y							
t_{PHL}	3004 Fig. 5	124	↓	↓	↓	↓	↓		GND		↓	↓	↓	GND			1Y	15	300	23	450	15	300	
		125		GND	↓												1Y							
		126			GND	↓											1Y							
		127				GND	↓										1Y							
		128					GND				↓				↓		2Y							
		129									GND	↓					2Y							
		130										GND	↓				2Y							
		131											GND	↓			2Y							
t_{TLH}		132	↓	↓	↓	↓	↓		GND		↓	↓	↓	GND			1Y	21	420	31	615	21	420	
		133		GND	↓												1Y							
		134			GND	↓											1Y							
		135				GND	↓										1Y							
		136					GND				↓				↓		2Y							
		137									GND	↓					2Y							
		138									GND	GND	↓				2Y							
		139									GND	GND	GND	↓			2Y							

See notes at end of device type 04.

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TABLE III. Group A inspection for device type 54.

Symbol	MIL-STD-883 method	Cases A, C, D X, Y Test No.	TERMINAL CONDITIONS 1/ 1 1A	2 1B	3 2A	4 2B	5 2C	6 2Y	7 V_{SS}	8 1C	9 1Y	10 3Y	11 3A	12 3B	13 3C	14 V_{DD}	Meas terminal	Test limits Subgroup 1 T_A = 25°C Min	Max	Subgroup 2 T_A = 125°C Min	Max	Subgroup 3 T_A = -55°C Min	Max	Units
V_{IC} (pos)		1	1 mA													GND	1A		1.5					Vdc
		2		1 mA													1B							
		3								1 mA							1C							
		4			1 mA												2A							
		5				1 mA											2B							
		6					1 mA										2C							
		7											1 mA				3A							
		8												1 mA			3B							
		9													1 mA		3C							
V_{IC} (neg)		10	-1 mA						GND								1A		-6					
		11		-1 mA													1B							
		12								-1 mA							1C							
		13			-1 mA												2A							
		14				-1 mA											2B							
		15					-1 mA										2C							
		16											-1 mA				3A							
		17												-1 mA			3B							
		18													-1 mA		3C							
I_{ss} 2/	3005	19	18 V	GND	18 V	GND	GND			GND			18 V	GND	GND	18 V	V_{DD}		-25		-750			nA
		20	GND	18 V	GND	18 V	GND			GND			GND	18 V	GND		V_{SS}							
		21		GND	GND	GND	18 V			18 V				GND	18 V		V_{SS}							
		22		GND	GND	GND	GND			GND				GND	GND		V_{SS}							
V_{OHS}	3006	23	GND	GND	15 V	15 V	15 V			GND			15 V	15 V	15 V	15 V	1Y	14.95		14.95		14.95		Vdc
		24	15 V	15 V	GND	GND	GND			15 V			15 V	15 V	15 V		2Y	14.95		14.95		14.95		
		25	15 V	15 V	15 V	15 V	15 V						GND	GND	GND		3Y	14.95		14.95		14.95		
V_{OLS}	3007	26	15	GND	GND	GND	GND			GND							1Y		0.05		0.05		0.05	
		27	GND	15 V	GND					GND							1Y							
		28		GND	GND					15 V							1Y							
		29			15 V					GND							2Y							
		30			GND	15 V											2Y							

See notes at end of table.

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TABLE III. Group A inspection for device type 54 – Continued.

Symbol	MIL-STD-883 method	Cases A, C, D X, Y	TERMINAL CONDITIONS 1/														Meas terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test No.	1A	1B	2A	2B	2C	2Y	V_{SS}	1C	1Y	3Y	3A	3B	3C	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{OLS}	3007	31	GND	GND	GND	GND	15 V		GND	GND			GND	GND	GND	15 V	2Y		0.05		0.05		0.05	Vdc
		32					GND						15 V	GND			3Y							
		33											GND	15 V			3Y							
		34											GND	GND	15 V		3Y							
V_{IH1}		35	3.5 V	1.5 V	1.5 V	1.5 V	1.5 V			1.5 V			1.5 V	1.5 V	1.5 V	5 V	1Y		0.5		0.5		0.5	
		36	1.5 V	3.5 V	1.5 V					1.5 V							1Y							
		37		1.5 V	1.5 V					3.5 V							1Y							
		38			3.5 V					1.5 V							2Y							
		39			1.5 V	3.5 V											2Y							
		40				1.5 V	3.5 V										2Y							
		41					1.5 V						3.5 V				3Y							
		42											1.5 V	3.5 V			3Y							
		43											1.5 V	1.5 V	3.5 V		3Y							
V_{IH2}		44	7.0 V	3.0 V	3.0 V	3.0 V	3.0 V			3.0 V			3.0 V	3.0 V	3.0 V	10 V	1Y		1.0		1.0		1.0	
		45	3.0 V	7.0 V						3.0 V							1Y							
		46		3.0 V						7.0 V							1Y							
		47			7.0 V					3.0 V							2Y							
		48			3.0 V	7.0 V											2Y							
		49				3.0 V	7.0 V										2Y							
		50					3.0 V						7.0 V				3Y							
		51					3.0 V						3.0 V	7.0 V			3Y							
		52					3.0 V						3.0 V	3.0 V	7.0 V		3Y							
V_{IH3}		53	11 V	4.0 V	4.0 V	4.0 V	4.0 V			4.0 V			4.0 V	4.0 V	4.0 V	15 V	1Y		1.5		1.5		1.5	
		54	4.0 V	11 V	4.0 V					4.0 V							1Y							
		55		4.0 V	4.0 V					11 V							1Y							
		56			11 V					4.0 V							2Y							
		57			4.0 V	11 V											2Y							
		58				4.0 V	11 V										2Y							
		59					4.0 V						11 V				3Y							
		60					4.0 V						4.0 V	11 V			3Y							
		61					4.0 V						4.0 V	4.0 V	11 V		3Y							

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TABLE III. Group A inspection for device type 54 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	TERMINAL CONDITIONS 1/														Meas terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test No.	1A	1B	2A	2B	2C	2Y	V_{EE}	1C	1Y	3Y	3A	3B	3C	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IL1}		62	1.5 V	1.5 V	5 V	5 V	5 V		GND	1.5 V			5 V	5 V	5 V	5 V	1Y	4.5		4.5		4.5		Vdc
↓		63	5 V	5 V	1.5 V	1.5 V	1.5 V		↓	5 V			5 V	5 V	5 V	5 V	2Y	↓		↓		↓		↓
		64	5 V	5 V	5 V	5 V	5 V			5 V			1.5 V	1.5 V	1.5 V	5 V	3Y							
V_{IL2}		65	3.0 V	3.0 V	10 V	10 V	10 V			3.0 V			10 V	10 V	10 V	10 V	1Y	9.0		9.0		9.0		Vdc
V_{IL2}		66	10 V	10 V	3.0 V	3.0 V	3.0 V			10 V			10 V	10 V	10 V	10 V	2Y	↓		↓		↓		↓
V_{IL2}		67	10 V	10 V	10 V	10 V	10 V			10 V			3.0 V	3.0 V	3.0 V	10 V	3Y							
V_{IL3}		68	4.0 V	4.0 V	15 V	15 V	15 V			4.0 V			15 V	15 V	15 V	15 V	1Y	13.5		13.5		13.5		Vdc
V_{IL3}		69	15 V	15 V	4.0 V	4.0 V	4.0 V			15 V			15 V	15 V	15 V	15 V	2Y	↓		↓		↓		↓
V_{IL3}		70	15 V	15 V	15 V	15 V	15 V			15 V			4.0 V	4.0 V	4.0 V	15 V	3Y							
I_{IL1}		71	5 V	GND	GND	GND	GND			GND	0.4 V		GND	GND	GND	5 V	1Y	0.51		0.36		0.64		mA
↓		72	GND	5 V	↓	↓	↓			GND	0.4 V		↓	↓	↓	↓	1Y	↓		↓		↓		↓
		73		GND						5V	0.4 V						1Y							
		74			5 V	↓		0.4 V		GND							2Y							
		75			GND	5 V	↓	0.4 V									2Y							
		76				GND	5 V	0.4 V					↓				2Y							
		77					GND					0.4 V	5 V	↓			3Y							
		78										0.4 V	GND	5 V	↓		3Y							
		79	↓	↓	↓	↓	↓	↓	↓	↓	↓	0.4 V	GND	GND	5 V	↓	3Y	↓		↓		↓		↓
I_{IL2}		80	15 V								1.5 V				GND	15 V	1Y	3.4		2.4		4.2		
↓		81	GND	15 V						↓	1.5 V				↓	↓	1Y	↓		↓		↓		↓
		82		GND	↓					15 V	1.5 V						1Y							
		83			15 V	↓		1.5 V		GND							2Y							
		84			GND	15 V	↓	1.5 V									2Y							
		85				GND	15 V	1.5 V					↓				2Y							
		86					GND					1.5 V	15 V	↓			3Y							
		87					GND					1.5 V	GND	15 V	↓		3Y							
		88	↓	↓	↓	↓	GND	↓	↓	↓	↓	1.5 V	GND	GND	15 V	↓	3Y	↓		↓		↓		↓

See notes at end of table.

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TABLE III. Group A inspection for device type 54. - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D X, Y	TERMINAL CONDITIONS 1/														Meas terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test No.	1A	1B	2A	2B	2C	2Y	V_{EE}	2C	1Y	3Y	3A	3B	3C	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{OHX}		89	GND	GND	5 V	5 V	5 V		GND	GND	4.6 V		5 V	5 V	5 V	5 V	1Y	-0.51		-0.36		-0.64		mA
		90	5 V	5 V	GND	GND	GND	4.6 V		5 V			5 V	5 V	5 V	5 V	2Y	-0.51		-0.36		-0.64		
		91	5 V	5 V	5 V	5 V	5 V			5 V		4.6 V	GND	GND	GND	5 V	3Y	-0.51		-0.36		-0.64		
I_{OHY}		92	GND	GND	10 V	10 V	10 V			GND	13.5 V		10 V	10 V	10 V	15 V	1Y	-3.4		-2.4		-4.2		
		93	10 V	10 V	GND	GND	GND	13.5 V		10 V			10 V	10 V	10 V	15 V	2Y	-3.4		-2.4		-4.2		
		94	10 V	10 V	10 V	10 V	10 V			10 V		13.5 V	GND	GND	GND	15 V	3Y	-3.4		-2.5		-4.2		
I_{IL} 2/	3009	95	GND	GND	GND	GND	GND		GND	GND			GND	GND	GND	18.0 V	All inputs together		-9.0					nA
		96	GND	GND	GND	GND	GND		GND	GND			GND	GND	GND	18.0 V	1A		-1.0		-45			
		97															1B							
		98															1C							
		99															2A							
		100															2B							
		101															2C							
		102															3A							
		103															3B							
		104															3C							
I_{IH} 2/	3010	105	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V		GND	18.0 V			18.0 V	18.0 V	18.0 V	18.0 V	All inputs together		9.0					
		106	18.0 V	GND	GND	GND	GND		GND	GND			GND	GND	GND	18.0 V	1A		+1.0		45			
		107	GND	18.0 V	GND					GND							1B							
		108		GND	GND					18.0 V							1C							
		109			18.0 V					GND							2A							
		110			GND	18.0 V											2B							
		111				GND	18.0 V										2C							
		112					GND						18.0 V				3A							
		113											GND	18.0 V			3B							
		114											GND	GND	18.0 V		3C							

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TABLE III. Group A inspection for device type 54 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y Test No.	TERMINAL CONDITIONS 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Test limits Subgroup 4 T_A = 25°C Min	Max	Subgroup 9 T_A = 25°C Min	Max	Subgroup 10 T_A = 125°C Min	Max	Subgroup 11 T_A = -55°C Min	Max	Units
C_I	3012	115	1A	1B	2A	2B	2C	2Y	GND	1C	1Y	3Y	3A	3B	3C	V_{cc}	1A	12								pF
		116															1B									
		117															1C									
		118															2A									
		119															2B									
		120															2C									
		121															3A									
		122															3B									
		123															3C									
t_{PHL}	3012 Fig. 5	124		GND		GND		GND	GND	GND		GND	GND	GND		3.0 V	1A to 1Y	11	210	11	210	16	315	11	210	ns
		125	GND		GND		GND			GND		GND	GND	GND			1B to 1Y									
		126															1C to 1Y									
		127															2A to 2Y									
		128															2B to 2Y									
		129															2C to 2Y									
		130															3A to 3Y									
		131															3B to 3Y									
		132															3C to 3Y									
t_{PLH}	3003 Fig. 5	133		GND		GND		GND	GND	GND		GND	GND	GND		3.0 V	1A to 1Y	11	210	11	210	16	315	11	210	ns
		134	GND		GND		GND			GND		GND	GND	GND			1B to 1Y									
		135															1C to 1Y									
		136															2A to 2Y									
		137															2B to 2Y									
		138															2C to 2Y									
		139															3A to 3Y									
		140															3B to 3Y									
		141															3C to 3Y									

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TABLE III. Group inspection for device type 54 – Continued.

Symbol	MIL-STD-883 method	Cases A, C, D X, Y Test No.	TERMINAL CONDITIONS 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Test limits Subgroup 9 T_A = 25°C Min	Max	Subgroup 10 T_A = 125°C Min	Max	Subgroup 11 T_A = −55°C Min	Max	Units
			1A	1B	2A	2B	2C	2Y	V_{cc}	1C	1Y	3Y	3A	3B	3C	V_{cc}								
t_{TLH}	3004 Fig. 5	142	5/	GND	GND	GND	GND		GND	GND	6/		GND	GND	GND	5.0 V	1Y	15	300	23	450	15	300	ns
		143	GND	5/						GND	6/						1Y							
		144		GND						5/	6/						1Y							
		145			5/			6/		GND							2Y							
		146			GND	5/		6/									2Y							
		147				GND	5/	6/									2Y							
		148					GND					6/	5/				3Y							
		149										6/	GND	5/			3Y							
		150										6/	GND	GND	5/		3Y							
t_{THL}		151	5/	GND	GND	GND	GND		GND		6/	GND	GND	GND	GND	5.0 V	1Y	21	410	31	615	21	410	
		152	GND	5/							6/						1Y							
		153		GND						5/	6/						1Y							
		154			5/			6/		GND							2Y							
		155			GND	5/		6/									2Y							
		156				GND	5/	6/									2Y							
		157					GND					6/	5/				3Y							
		158					GND					6/	GND	5/			3Y							
		159					GND					6/	GND	GND	5/		3Y							

NOTES:

1/ Input pins not designated may be tied to V_{CC} or GND or may be left open provided they do not influence the outcome of the measurement. Output pins not designated may be tied to the loads or may be left open provided they do not influence the outcome of the measurement, except as follows: V_{IC} (pos) tests; the V_{IN} terminals shall be open. V_{IC} (neg) tests; the V_{IN} terminal shall be open. I_{OS} tests; the output terminals shall be open.

2/ When performing quiescent supply current measurements (I_{CC}), the meter shall be placed so that all currents flow thru the meter.

3/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

4/ Capacitance bridge between measured terminal and V_{CC}; frequency = 1 MHz.

5/ Pulse conditions – See figure 5

6/ Loading conditions – See figure 5.

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b. A special subgroup shall be added using an LTPD of 15 for classes S and B, and shall be performed on each inspection lot for initially qualified device types 01 through 04, and measured only for initial qualification and after process or design changes for initially qualified device types 51 through 54. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

c. End-point electrical parameters shall be as specified in table II herein and shall consist only of those subgroups specified in table IIa of test method 5005 of MIL-STD-883, and table II herein. Delta limits shall apply only to subgroup 5 of group B inspections and shall consist of tests specified in table IV herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D and as specified in 4.5.2 herein and as shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration, 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices, and shall consist of the tests, conditions, and limits as specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.1). When group E testing is performed it shall be in accordance with table V of method 5005 of MIL-STD-883 and 4.5.5 herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. Unless otherwise specified, all voltages given are referenced to the microcircuit V_{SS} terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

TABLE IV. Delta limits at 25°C.

Parameter 1/	Device types	
	01-04	51-54
I_{SS}	±10 nA	±10 nA
V_{OL1}	±.04 V	---
V_{OH1}	±.08 V	---
I_{OL1}	---	±15%
I_{OH1}	---	±15%

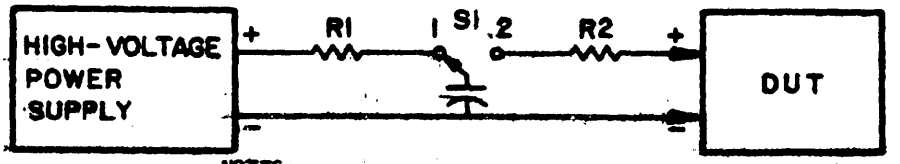
1/ Each of the above parameters shall be recorded before and after the required burn-in or life tests to determine deltas (Δ).

4.5.2 Burn-in and life test cool down procedures. When burn-in and life tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to a temperature of 25°C ±3°C; then, electrical parameter end-point measurements shall be performed.

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4.5.3 High voltage (V_{ZAP}) test of input protection circuits. All input terminals (up to a maximum of 4) of the DUT shall be subjected to a voltage pulse from a 100 pF source charged to 400 V. This destructive test shall be conducted as follows using the test circuit shown on figure 6.

a. Measure I_{IL} and I_{IH} at the inputs selected, as stated above, at 25°C. The test limit for each input tested shall be ±10 nA at the specified V_{DD}. Measure I_{SS} on the DUT at 25°C. The test limit for this measurement shall be increased a maximum of 20 percent of the specified I_{SS} table III limit at the specified V_{DD}.



NOTES:
1. V_{ZAP} = 400 V charge on C1.
2. 1 MΩ ≤ R1 ≤ 50 MΩ.
3. R2 = 1.5 kΩ.
4. C1 = 100 pF.
5. S1 = Hg-wetter "bounceless" relay.

FIGURE 6. High voltage (V_{ZAP}) test circuit.

b. V_{ZAP} is applied to DUT in the following modes (see table V) by changing C1 to V_{ZAP} with S1 in position 1 and then switching to position 2.

TABLE V. Modes for high voltage test.

Mode	+ Terminal	- Terminal
1	V_{DD}	Input
2	Input	V_{SS}
3	Input	Associated output

c. Within 24 hours repeat the I_{SS}, I_{IL}, and I_{IH} measurements on the same terminals as performed above. If a DUT exhibits leakage currents in excess of the specified limits after the V_{ZAP} test, it shall be classified as a failure.

4.5.4 Quiescent supply current (I_{SS}) test. When performing quiescent supply current measurements, the meter shall be placed so that all currents flow through the meter.

4.5.5 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in table V of method 5005 of MIL-STD-883 and herein:

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (group A subgroup 1 at 25°C) and also be subjected to the threshold voltage test in table VIII in order to calculate the delta threshold (ΔV_T) after irradiation.

b. The devices shall be subjected to a total radiation dose as specified in MIL-M-38510 for the RHA level being tested, and meet the end-point electrical parameters as defined in table VI at 25°C, after exposure. The start and completion of the end-point electrical parameter measurements shall not exceed 2 hours following irradiation.

c. Threshold voltage test circuit conditions shall be as specified in table VIII and figure 7. In situ and remote testing, the tests shall be performed with the devices biased in accordance with table VII and bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.

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d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required, then an equivalent truth table test shall be performed.

TABLE VI. Radiation hardening end-point electrical parameters at 25°C.

Parameter	All device types	V_{DD}	
		Device types 01, 04	Device types 51, 54
V_{TN}	0.3 V min	10 V	10 V
V_{TP}	2.8 V max	10 V	10 V
ΔV_T	1.4 V	10 V	10 V
I_{SS}	100 x max limit	15 V	18 V
t_{PLH}	1.35 x max limit	5 V	5 V
t_{PHL}	1.35 x max limit	5 V	5 V

TABLE VII. Bias during exposure to radiation.

Device type	Pin connections		
	V_{DD} = 10 V dc (through a 30- to 60-kΩ resistor)	V_{SS} = GND	V_{DD} = 10 V dc
01, 51	3, 4, 5, 8, 11, 12, 13	7	14
02, 52	1, 2, 5, 6, 8, 9, 12, 13	7	14
03, 53	2, 3, 4, 5, 9, 10, 11, 12	7	14
04, 54	1, 2, 3, 4, 5, 8, 11, 12, 13	7	14

Pins not designated are open or connected to 10 V dc through a 30- to 60-kilohm resistor.

TABLE VIII. Threshold voltage circuit conditions.

Device type	GND	10 V	V_{TN} measured at		GND	-10 V	V_{TP} measured at	
			20 µA supply	10 µA supply			20 µA supply	10 µA supply
01,51	3	14		1,2,4,5, 7,8,11-13	3	1,2,4,5, 7,8,11-13		14
02,52	1	14		2,5-9,12, 13	1	2,5-9, 12,13		14
03,53	2	14		3-5,7, 9-12	2	3-5,7, 9-12		14
04,54	3	14		1,2,4,5, 7,8,11-13	3	1,2,4,5, 7,8,11-13		14

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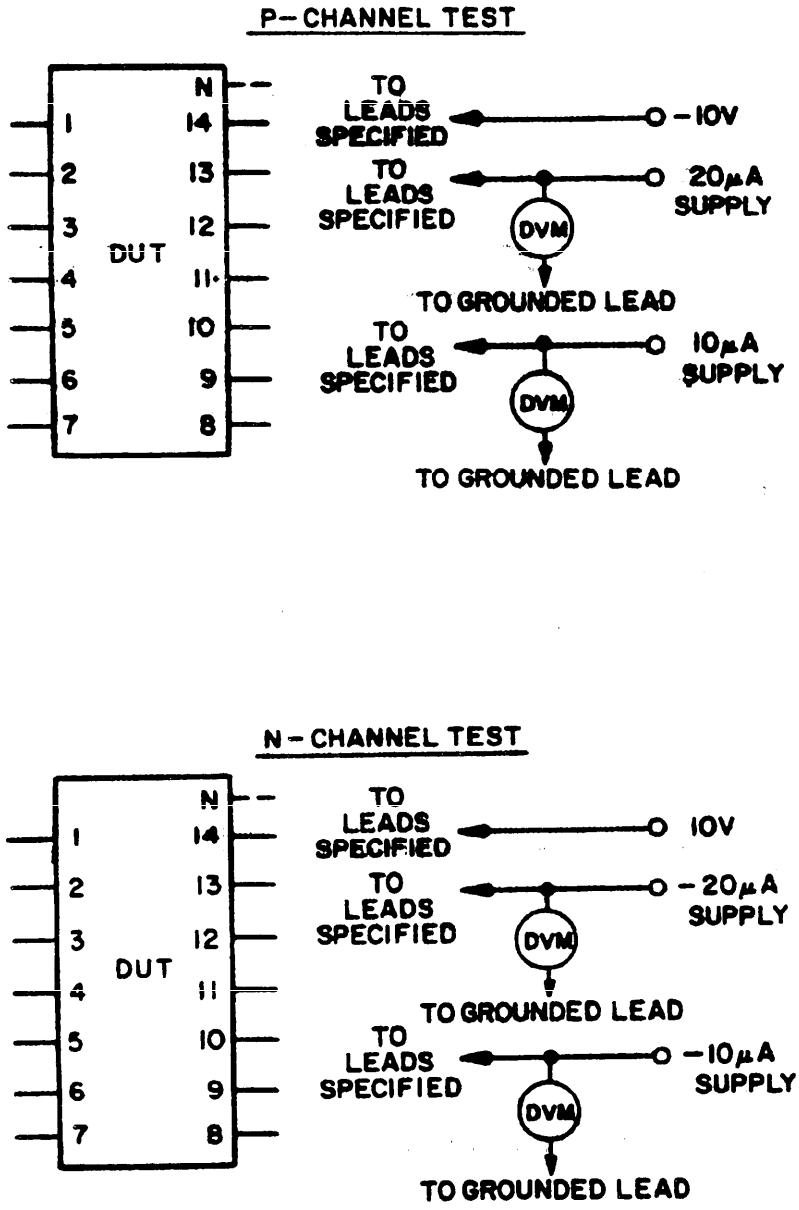


FIGURE 7. Threshold-voltage test circuit conditions.

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4.6 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, shall be supplied.

a. Attributes data for all screening tests (see 4.2 and variables data for all static burn-in, dynamic burn-in, and steady state life tests (see 3.5).

b. A copy of each radiograph (see 4.2).

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parameters data (see 4.2c).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510. These devices require electrostatic protection.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logisitic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by, or direct shipment to the Government.

h. Requirements for "JAN" marking.

i. Requirements for total dose radiation testing (see 3.6.1 and 4.5.5), if applicable.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

C_I	Input terminal-to-V_{SS} capacitance.
GND	Ground zero voltage potential.
I_{SS}	Quiescent supply current.
T_A	Free air temperature.
V_{DD}	Positive supply voltage.
$V_{IC(pos)}$	Positive clamping input to V_{DD}.
$V_{IC(neg)}$	Negative clamping input to V_{SS}.
V_{SS}	Negative supply voltage.
V_{ZAP}	Input test voltage.

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Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	4000A
02	4001A
03	4002A
04	4025A
51	4000B
52	4001B
53	4002B
54	4025B

6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static buildup. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - AR, SM
Navy - AS, CG, MC, OS, SH

Agent:
DLA - ES

Preparing activity:
NASA - NA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/52D	MICROCIRCUITS, DIGITAL, CMOS, NOR GATES, MONOLITHIC SILICON, POSITIVE LOGIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
3b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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